Physics-constrained generative models of a parametric MHD dynamo

M2 internship report · race pilot, team claude · a comparative study of constraints imposed by design, loss and sampling

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Status of this document This report is the deliverable of team claude in the race pilot of this repository (`race/DELIVERABLES.md`). It is written in the form of the M2 internship report that the MilaDy offer asks for, by an AI agent working alone for one session, and it reports one experiment, E4, designed, run and reviewed during that session. Nothing in it has been read or validated by the supervisors of the internship, and no data of the MilaDy consortium was used.

Three statuses are kept apart throughout. What is **measured** comes from result files written by committed scripts of this repository (`experiments/results/`, summarised in `experiments/results/digest.md`); every table of numbers below is generated from those files by `experiments/e4_report_tables.py`. What is **published** is cited with a reference taken from the verified source files `docs/specs/sources/`. What is **conjectured** is said to be so. The system studied is small (a 16^3 grid, 665 Fourier modes, Reynolds number 10): the conclusions are about the mechanisms, not about turbulent dynamos.

The HTML version of this report and an interactive explorer of the results are published with the site of the repository, under `teams/claude/report/` and `teams/claude/explorer/`.

Abstract

The MilaDy internship offer asks for generative models of parameter-dependent 3D magneto-hydrodynamic fields in which physical constraints are imposed through model design, training losses or sampling procedures, and for a comparative study of these strategies. This report carries out that comparison on a system small enough to be simulated and learned many times: a helically forced dynamo in a periodic box (16^3 grid, 665 Fourier modes, $Re = 10$), conditioned on the magnetic diffusivity η, with training, validation and held-out values of η, five model seeds per variant, and hypotheses registered with a refutation criterion before each computation. Thirteen variants of conditional flow matching on Fourier modes are compared with a diffusion model and a Gaussian baseline on an energy distance to the reference law relative to its noise floor, the divergence, the total and magnetic energy balances, spectra and helicity, the number of magnetic null points, a restart test in the solver and a polarity test.

All results are **measured** in this repository. Zero divergence is exact by design (vector potentials) or by projection, while a divergence penalty fails to enforce it. These mechanisms do not decide the quality of the law: every first-generation model is more than a hundred noise floors from the reference, and the plain networks fit the data worse than a trivial Gaussian velocity field. Giving the velocity field an explicit mode-wise affine term reduces the distance by a factor of about twenty, to nine floors. A Gaussian law per Fourier mode, without a network, is closer in energy distance (3.5 floors) but is not a dynamo: its mean magnetic production vanishes, it has three times too many magnetic null points, and its magnetic energy drifts when the solver restarts from its samples. The flow-matching models with the affine term restore the magnetic energy balance (residual 0.06 against 1) and halve that drift, at the price of less accurate spectra. A law-level stationarity loss on the energy identity and on the parameter-free identity $\mathbb{E}[\langle j \cdot B \rangle] = 0$ lowers the residual of the energy balance at parameter values without data; the residual of the second identity does not respond, and the loss has side effects on other metrics. Six of the eleven hypotheses of the first protocol were refuted and are reported as such. The code (twelve tested constraint mechanisms, a diffusion sampler, a null-point finder) is in the project toolbox.

1 Introduction

1.1 Context: the internship offer and the MilaDy project

The internship offer that this report answers is entitled *Physics-Constrained Generative AI Modelling of Parametric Dynamical Systems*. It belongs to the ANR project MilaDy, on machine-learning models of turbulent dynamos, and asks for generative models (diffusion or flow matching) of parameter-dependent three-dimensional fields in which physical constraints are imposed "through model design, training losses or sampling procedures". The models are to be evaluated with energy spectra, helicity, divergence control and the preservation of topology, and the expected outcome is a comparative study of the strategies together with code integrated into the project toolbox (`docs/specs/sources/supervisors.md`, §1.2).

A dynamo is a flow of an electrically conducting fluid that sustains its own magnetic field. The equations are those of magnetohydrodynamics (MHD): the Navier–Stokes equations with the Lorentz force, coupled to the induction equation. Two control parameters matter here, the kinetic Reynolds number Re and the magnetic Reynolds number Rm, which measures induction against magnetic diffusion. Simulating a dynamo at one parameter value is expensive, and the scientific questions (thresholds, saturation levels, statistics of the saturated state) are questions about how the statistically stationary state changes with the parameters. A conditional generative model addresses exactly that object: it learns the law of the snapshots of the stationary state as a function of the parameter and can then draw new snapshots at parameter values that were never simulated.

The difficulty, and the subject of the offer, is that a generated field must be physically admissible. The magnetic field must be divergence-free. The energy injected by the forcing must, on average, be dissipated. The symmetries of the equations must be those of the generated law. The topology of the magnetic field, for instance its null points, should be that of the reference. The published literature offers three places where such requirements can enter a generative model, which are the three places named by the offer: the design of the model, the training loss, and the sampling procedure.

1.2 Question and approach

The question of this report is narrow and practical: **for a parametric 3D MHD dynamo, which constraints should be imposed where, and what does each mechanism buy on the metrics the offer names?** The approach is a controlled comparison on a system small enough to be simulated, learned and evaluated many times: a helically forced dynamo in a periodic box, on a 16^3 grid, with the magnetic diffusivity η as the conditioning parameter (Figure 1.1). On this system the reference law can be sampled at will, every quantity can be computed to round-off, and each comparison can be repeated over five model seeds.

The work follows the protocol of the repository (`AGENTS.md`): each hypothesis is written in a register with a quantitative refutation criterion before the computation; each quoted comparison uses five seeds; a reviewing role, the `encadrant`, recomputes the result and gives the verdict;

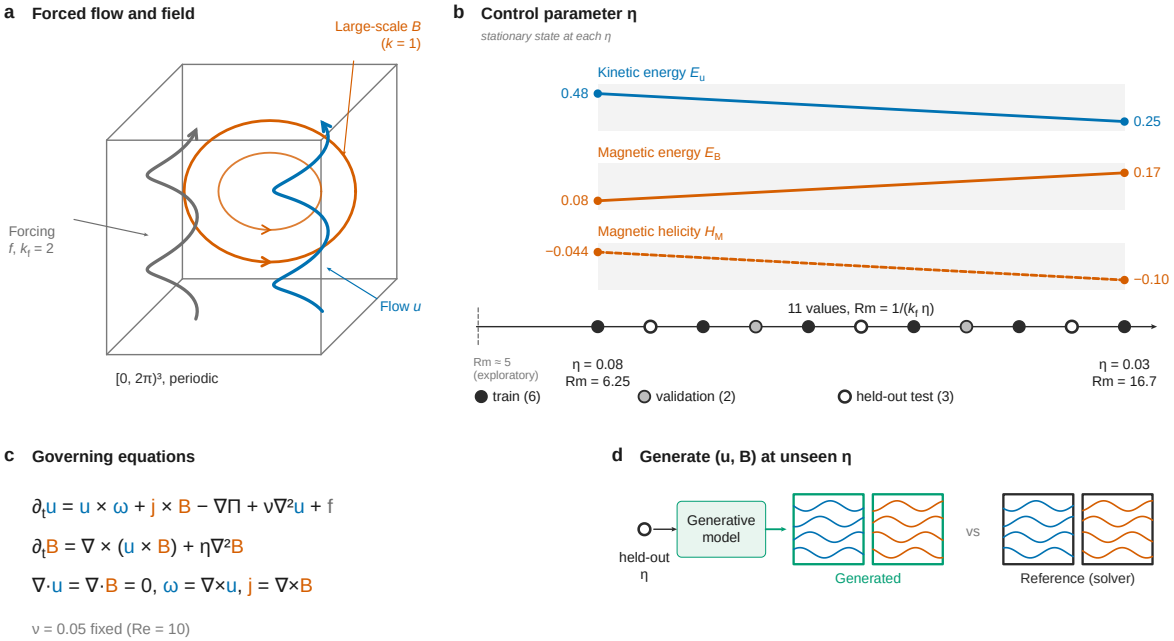


Figure 1.1: The parametric problem of experiment E4. (a) Periodic box with the helical ABC forcing at wavenumber $k_f = 2$, the flow it drives and the large-scale magnetic field at $k = 1$. (b) The eleven values of the magnetic diffusivity η, alternating between training, validation and held-out test values, with the trends of the kinetic energy, magnetic energy and magnetic helicity of the saturated state (end values measured, segments schematic). (c) The equations; the viscosity is fixed. (d) The task: draw snapshots (u, B) at a held-out η and compare their law with that of the solver. Schematic drawn with the `schematic-figure` skill; source in `figures/mhd-dynamo-problem/`.

negative results are kept. Twenty-one hypotheses were registered, eighteen of them in this experiment (H4 to H21).

1.3 Contributions

All of the following are measured in this repository, on the system described above.

1. **A data set and a metric suite for a parametric 3D dynamo** (Experiments): 11 values of η, 24 trajectories each, with a convergence check on finer grids, and metrics covering the four families of the offer (spectra, helicity, divergence, topology), the energy budgets, and two tests that go beyond statistics chosen in advance, a restart test in the solver and a polarity test.
2. **A comparison of constraint mechanisms by the place where they act** (Comparative study): by design (vector potentials, equivariance under $B \to -B$, an explicit affine term, an anchor on a Gaussian law), by the loss (divergence penalty, law-level stationarity identities) and by the sampling procedure (Leray projection, projected flow, exponential tilting, per-sample rescaling), against an unconstrained baseline, a diffusion model and a Gaussian baseline without a network.
3. **A diagnosis of why the first generators fail and a remedy** (Results): residual networks narrower than the latent dimension do worse than the trivial Gaussian velocity field; giving the velocity field its linear part explicitly reduces the distance to the reference law by a factor of about twenty.
4. **A stationarity identity without parameters**, $\mathbb{E}[\langle j \cdot B \rangle] = 0$, that follows from the magnetic-helicity budget and can be imposed at parameter values without data (Methods), and the measurement of what a law-level loss built on it and on the energy identity does and does not change.
5. **Code in the project toolbox** (Toolbox): twelve registered constraint mechanisms with one test each, a diffusion sampler, a null-point finder, and the tooling to run and account for GPU jobs.

The report also states what was refuted: six of the hypotheses registered before the computation were refuted and one was dropped as not discriminating (Discussion).

1.4 Outline

the section State of the art places the work in the literature. The section Methods defines the system, the generative model, each constraint mechanism and each metric. The section Experiments gives the data and the protocol. The section Results reports the measurements, the section Comparative study gathers them in one table with the verdict of each hypothesis, the section Toolbox describes the code, the section Discussion discusses limits and negative results, and the section Reproducibility lists the commands, commits, GPU jobs and costs.

2 State of the art

Everything in this section is **published** work, cited from the verified source files of the repository (`docs/specs/sources/literature.md`, `dynamo.md`, `data-metrics.md`); the full references are in the section References.

2.1 Diffusion and flow matching

Diffusion models learn to reverse a noising process, either as a discrete chain (Ho, Jain, and Abbeel 2020) or as a stochastic differential equation whose reverse drift involves the score of the noised law (Song et al. 2021). Flow matching (Lipman et al. 2023) and the closely related rectified flows (Liu, Gong, and Liu 2023) and stochastic interpolants (Albergo and Vanden-Eijnden 2023) learn instead the velocity field of an ordinary differential equation that transports a simple law to the data law; with a straight path between a noise sample and a data sample, the regression target is the difference of the two. The parameterisation of a diffusion model by a "velocity" $v = \alpha_t \varepsilon - \sigma_t x$ was introduced by Salimans and Ho (2022) (their Section 4) and is the one used for the diffusion baseline of this report. Both families extend to function spaces (Kerrigan, Migliorini, and Smyth 2024; Lim et al. 2025).

2.2 Generative models of flows and of PDE solutions

Generative models have been applied to turbulent flows to draw turbulent states directly, without rolling out a simulation (Lienen et al. 2024), to PDE solutions in function space with constraints built into the architecture (Wang et al. 2026), to reconstruction from sparse data (Shu, Li, and Barati Farimani 2023; Huang et al. 2024), and to latent representations of flows (Du et al. 2024). FunDiff (Wang et al. 2026) is the main reference of the offer: a generative model over function spaces, with a latent model trained by rectified flow and a decoder that can enforce constraints such as zero divergence through a stream function.

2.3 Three places for a physical constraint

By design. A field written as the curl of a potential is divergence-free whatever the potential; Richter-Powell, Lipman, and Chen (2022) build divergence-free networks on this principle, and FunDiff uses a stream function. Symmetries can also be built in: the flow of an equivariant vector field transports an invariant base law to an invariant law (Köhler, Klein, and Noé 2020).

By the training loss. Physics-informed diffusion models add the residual of the governing equation, evaluated on a denoised estimate, to the training loss (Bastek, Sun, and Kochmann 2025). The estimate is a conditional mean, so for a nonlinear residual the penalty does not vanish at the optimum (the Jensen gap discussed in that paper). Baldan et al. (2026) report a trade-off between the distributional accuracy and the physical accuracy of flow-matching models trained with a residual term.

By the sampling procedure. Constraints can be imposed during sampling by guidance with the gradient of a residual (Chung et al. 2023; Huang et al. 2024), by projection of each iterate onto the constraint set (Christopher, Baek, and Fioretto 2024), or by corrections interleaved with the integration of a flow-matching model, without retraining (Cheng et al. 2025; Utkarsh et al. 2025). For incompressible flows, the Leray projection has been used inside diffusion samplers (Genuist et al. 2026; Li et al. 2026).

2.4 Helical dynamos in periodic boxes

A flow forced with helicity at an intermediate scale in a periodic box generates a strong magnetic field at the scale of the box, and the time needed to build it grows with Rm (Brandenburg 2001); that paper attributes the underlying inverse cascade of magnetic helicity to Pouquet, Frisch, and Léorat (1976). The magnetic helicity $\langle A \cdot B \rangle$ is an invariant of ideal MHD (Woltjer 1958) and evolves, in a periodic domain, as $\mathrm{d}\langle A \cdot B \rangle / \mathrm{d}t = -2\eta \langle j \cdot B \rangle$ [the review of Brandenburg and Subramanian (2005), their Section 3.4, as recorded in the source file; in this work the identity is also checked numerically against the solver by a test]. The forcing used here is the ABC (Arnold–Beltrami–Childress) flow, whose kinematic dynamo properties have been studied in detail (Alexakis 2011; and Galloway and Frisch 1986, known here from its bibliographic record only). Dealiased pseudo-spectral codes are a standard tool for such periodic boxes (for instance Ponty et al. 2005).

2.5 Metrics

The energy distance of Székely and Rizzo (2013) is a distance between two laws computable from samples; it is zero only when the laws are equal. Magnetic null points, where $B = 0$, organise the topology of a magnetic field (Parnell et al. 1996); the trilinear method of Haynes and Parnell (2007) locates them on a grid.

2.6 What was not found

A literature search made for this repository on 6 November 2026 and extended on 12 November found no generative model trained on dynamo simulations, no use of the magnetic-helicity budget as a constraint of a generative model, no use of null-point counts as an evaluation metric of a generative model, and no restart test of generated MHD snapshots. These are negative statements: an absence in a search is not a proof of novelty, and the source files mark them as unconfirmed.

The repository's own theory brief (`docs/specs/theory-brief.md`) gives the propositions used below. Its Propositions 1 to 7 are elementary and probably known in other forms; in particular the use of weak stationarity identities as the loss of a sample generator is published (Cai et al. 2026), and the present work only claims their use for a field PDE, at the level of the law of a parametric conditional model.

3 Methods

3.1 The physical system

The fields are the velocity u and the magnetic field B (in Alfvén units) of an incompressible conducting fluid in the periodic box $[0, 2\pi)^3$:

$$\partial_t u = u \times \omega + j \times B - \nabla \Pi + \nu \nabla^2 u + f, \qquad \partial_t B = \nabla \times (u \times B) + \eta \nabla^2 B, \qquad \nabla \cdot u = \nabla \cdot B = 0, \tag{3.1}$$

with the vorticity $\omega = \nabla \times u$, the current density $j = \nabla \times B$ and the total pressure Π. The forcing is a constant ABC force at wavenumber $k_f = 2$,

$$f = \nu k_f^2 \, (\sin k_f z + \cos k_f y, \ \sin k_f x + \cos k_f z, \ \sin k_f y + \cos k_f x),$$

which is helical ($\nabla \times f = k_f f$) and whose amplitude makes the unit ABC flow the laminar solution without a magnetic field. The viscosity is fixed, $\nu = 0.05$, so that $Re = 1/(k_f \nu) = 10$. The control parameter is the magnetic diffusivity η, between 0.03 and 0.08, that is $Rm = 1/(k_f \eta)$ between 6.2 and 16.7.

The solver is the pseudo-spectral code of the toolbox (`milady.sim.mhd3d`) on a 16^3 grid with the 2/3 rule: the Fourier modes with $|m_i| \le 5$ in each direction are kept. Counting one mode for each pair $(m, -m)$ of a real field, that is $(11^3 - 1)/2 = 665$ independent complex modes.

3.2 Stationarity identities

A statistically stationary state has an invariant law μ_η. For any smooth observable φ of the state, the mean of its time derivative under μ_η is zero (Proposition 5 of the theory brief). Three quadratic observables give identities that involve only quantities computable on one snapshot (Figure 3.1):

$$\begin{aligned} \text{energy:} \quad & \mathbb{E}\big[\langle f \cdot u \rangle - \nu \langle |\omega|^2 \rangle - \eta \langle |j|^2 \rangle\big] = 0, \\ \text{magnetic helicity:} \quad & \mathbb{E}\big[\langle j \cdot B \rangle\big] = 0, \\ \text{cross helicity:} \quad & \mathbb{E}\big[\langle f \cdot B \rangle - (\nu + \eta) \langle \omega \cdot j \rangle\big] = 0, \end{aligned} \tag{3.2}$$

where $\langle \cdot \rangle$ is the spatial mean and $\mathbb{E}$ the mean over the law. The first is the balance between injection and dissipation. The second follows from the magnetic-helicity budget quoted above: in a periodic box the magnetic helicity changes only through $-2\eta \langle j \cdot B \rangle$, so the mean current helicity of a stationary state vanishes. It contains no parameter and no forcing term: it holds at every η, including values without data. The third is odd in B and is satisfied trivially by a law that is invariant under $B \to -B$.

A fourth identity is used as a metric and not imposed: the magnetic energy budget, $\mathbb{E}[P - \eta\langle|j|^2\rangle] = 0$ with the production $P = \langle j \cdot (u \times B)\rangle$, the balance between the work of the flow on the field (the dynamo term) and the Ohmic dissipation. P is cubic in the fields: it is a sum over triads of wavevectors $k+p+q=0$ of third-order moments $\mathbb{E}[\hat{u}(k)\hat{B}(p)\hat{B}(q)]$. For a Gaussian law with zero mean magnetic field these moments reduce exactly to $\mathbb{E}[\hat{u}(k)]\,\mathrm{Cov}(\hat{B}(p), \hat{B}(q))$: the mean production of a Gaussian law is the contraction of its mean flow with the covariance of the magnetic field between different Fourier modes. The constant forcing gives this system a strong mean flow at k_f and statistics that are not homogeneous, so that covariance exists. A law whose Fourier modes are independent has none of it and therefore almost no mean production, whatever its spectra; a Gaussian law that keeps the covariance between modes can have the right production. This elementary remark separates the two Gaussian baselines used below.

These are constraints on the law, not on each sample: the injection and the dissipation of one snapshot differ, and their difference is the instantaneous $\mathrm{d}E/\mathrm{d}t$. Forcing each sample to satisfy the balance sets the variance of $\mathrm{d}E/\mathrm{d}t$ to zero and cannot give the right law (Corollary 6 of the theory brief).

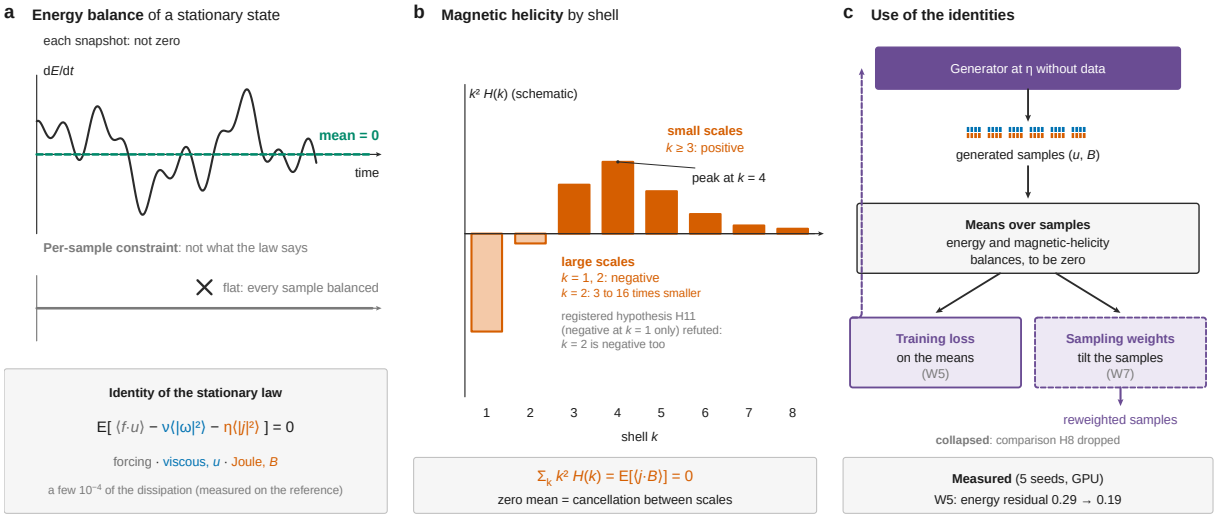


Figure 3.1: The three stationarity identities used in E4. Each is the mean, over the stationary law, of the time derivative of a quadratic invariant of ideal MHD; each term is computable on a single snapshot. The magnetic-helicity identity has no parameter in it. Schematic; source in `figures/stationarity-identities/`.

3.3 Latent representation and generative model

A snapshot (u, B) is transformed to Fourier space and restricted to the 665 modes of the solver. Two representations are used (Figure 3.2).

Free components. The three complex components of $\hat{u}$ and $\hat{B}$ at each mode: $665 \times 12 = 7980$ real numbers. Nothing forces $k \cdot \hat{u} = 0$ or $k \cdot \hat{B} = 0$.

Vector potentials. The potentials in Coulomb gauge, $\hat{A} = \mathrm{i}k \times \hat{B}/|k|^2$ and the same for u, written in the Craya–Herring basis (e_1, e_2) of the plane perpendicular to k: two complex numbers per mode and per field, $665 \times 8 = 5320$ real numbers. Decoding, $\hat{B} = \mathrm{i}k \times \hat{A}$, gives a divergence-free field whatever the latent (Proposition 1 of the theory brief). Because the modes are exactly

those of the solver, every quadratic quantity of Equation 3.2 computed from the latent is the one the solver would compute.

Both latents are normalised mode by mode. The generative model is conditional flow matching on the straight path (Lipman et al. 2023): with $x_0 \sim \mathcal{N}(0, I)$, a data latent x_1 at the condition c (the normalised $\log \eta$) and $x_t = (1-t)x_0 + tx_1$, a network $v_\theta(x, t, c)$ is trained by regression,

$$\mathcal{J}(\theta) = \mathbb{E}\left\|v_\theta(x_t, t, c) - (x_1 - x_0)\right\|^2, \tag{3.3}$$

and a sample is obtained by integrating $\mathrm{d}x/\mathrm{d}t = v_\theta(x, t, c)$ from $t = 0$ to 1 with Heun's scheme (50 steps). The network is a multilayer perceptron acting on the flattened latent: a linear map from the latent to a hidden space of width 512, four residual blocks in that space, and a linear map back to the latent, with Fourier embeddings of t and c.

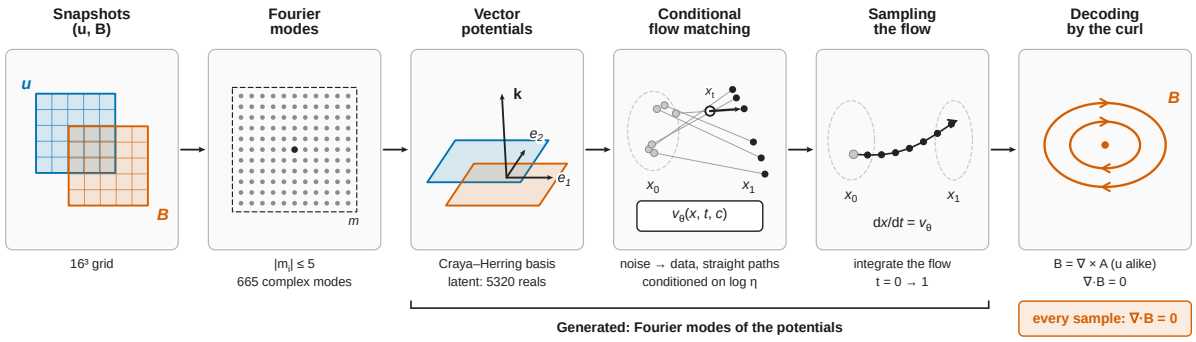


Figure 3.2: The generative pipeline: snapshots, Fourier modes of the solver, vector potentials in the Craya–Herring basis, conditional flow matching, sampling, decoding. Every decoded sample is divergence-free. Schematic; source in `figures/generative-pipeline/`.

3.4 Constraint mechanisms

The mechanisms are grouped by the place where they act (Figure 3.3); the names W0 to W11 and G0 are those of the result files.

Baseline (W0). Flow matching on the free components, no constraint.

Model design.

- *Vector potentials* (W4): flow matching on the potential latent.
- *Equivariance under* $B \to -B$ (W4s). Equation 3.1 is invariant under $B \to -B$, so the stationary law is too (when it is unique). With S the map that changes the sign of the magnetic channels, the field $v_{\mathrm{sym}}(x) = \frac{1}{2}\left[v(x) + Sv(Sx)\right]$ is equivariant and the generated law is exactly invariant (Proposition 4 of the theory brief; Köhler, Klein, and Noé (2020)).
- *Explicit affine term* (W9): $v(x, t, c) = G(t, c)\, x + b(t, c) + \mathrm{network}(x, t, c)$, with one 8×8 matrix G and one offset b per Fourier mode, both read linearly from an embedding of (t, c) and initialised at zero. The reason for this term is given in Section 3.5. W9 also uses the smooth embedding of the condition of W4c (frequencies up to half a cycle over the range of $\log \eta$), so it differs from W4s in two ways; the matched control, the same network with the smooth embedding and no affine term, is the variant of width 512 of the width scan.

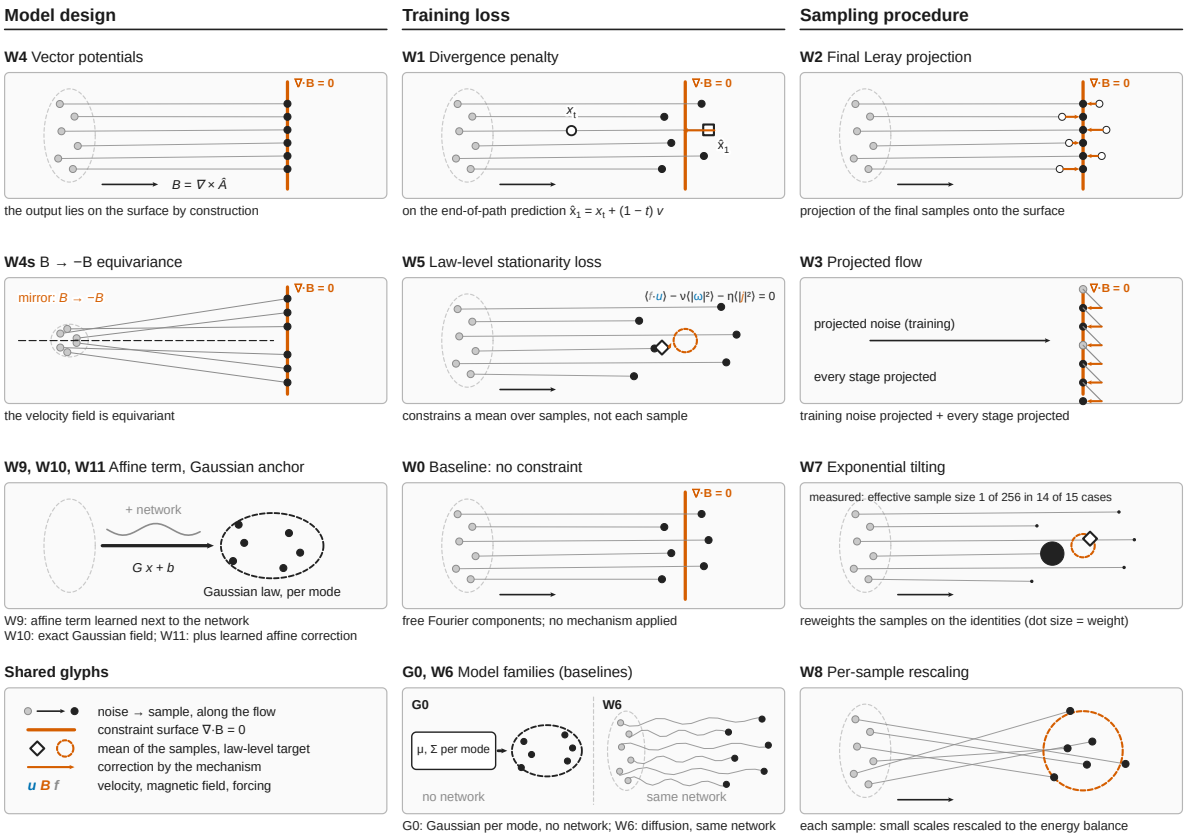


Figure 3.3: The constraint mechanisms compared in E4, by the place where they act: model design, training loss, sampling procedure. Dot positions are illustrative; no result is shown. Schematic; source in `figures/constraint-mechanisms/`.

- *Gaussian anchor* (W10) and *anchor with affine correction* (W11): $v = v_{\text{Gauss}}(x, t, c) +$ network, where v_{Gauss} is the exact flow-matching field of a mode-wise Gaussian law fitted to the data (Equation 3.4); W11 adds the learnable affine term of W9 to the correction.

Training loss.

- *Divergence penalty* (W1): on the free components, the loss Equation 3.3 plus 10 times the squared relative divergence of the end-of-path prediction $\hat{x}_1 = x_t + (1-t)v_\theta$. The constraint is linear, so the penalty vanishes at the optimal field and does not bias it (Proposition 3 of the theory brief).
- *Law-level stationarity loss* (W5): the equivariant potentials model is fine-tuned (400 iterations) with an additional term that draws, at four values of η taken at random in the whole range (with or without data), 16 samples through a differentiable 6-step integration, and penalises the squared normalised means of the energy and magnetic-helicity residuals of Equation 3.2. The square of a mean is estimated without bias by the product of the means of two half-batches; the naive square of one batch mean would add the variance of the residual and penalise the spread of the law. A control (W5_control) receives the same fine-tuning without the term.

Sampling procedure.

- *Final projection* (W2): Leray projection $\hat{f} - k\,(k \cdot \hat{f})/|k|^2$ of the samples of W0.
- *Projected flow* (W3): the noise is projected, the model is trained with projected noise, and every stage of Heun's scheme is projected, so that the whole path stays in the divergence-free subspace (Proposition 2 of the theory brief).
- *Exponential tilting* (W7): the samples of a trained model receive weights $w_i \propto \exp(\lambda \cdot g_i)$, where g_i are the per-sample residuals of the identities and λ is chosen so that the weighted means vanish; this is the law closest to the generated one, in relative entropy, that satisfies the identities (Proposition 7 of the theory brief; the existence and characterisation of this projection are due to Csiszár (1975)).
- *Per-sample rescaling* (W8): the modes above the forcing scale of each sample are multiplied by a factor that makes its own injection equal its own dissipation. This is the per-sample version of a law-level identity, included as the counter-example of Corollary 6.

Model families.

- *Diffusion* (W6): the same network and latent as W4, trained by v-prediction on a variance-preserving path (Salimans and Ho 2022) and sampled with the ancestral sampler (100 steps, the same number of network evaluations as 50 Heun steps).
- *Mode-wise Gaussian law* (G0): no network. For each training η, one mean and one 8×8 covariance per Fourier mode are estimated from the data, made invariant under $B \to -B$, and interpolated linearly in $\log \eta$. This law reproduces the second-order statistics of each mode (hence the spectra, the helicities and the mean budgets) and nothing else: it is the baseline that a generative model must beat to have learned anything non-Gaussian.

3.5 The linear part of the velocity field

Proposition (flow-matching field of a Gaussian law). Let $x_0 \sim \mathcal{N}(0, I)$ and $x_1 \sim \mathcal{N}(\mu, \Sigma)$ be independent and $x_t = (1-t)x_0 + tx_1$. The minimiser of Equation 3.3 is affine in x:

$$v_{\text{Gauss}}(x, t) = \mu + (t\,\Sigma - (1-t)\,I)(t^2\Sigma + (1-t)^2 I)^{-1}(x - t\mu). \tag{3.4}$$

Proof. The minimiser is the conditional mean $\mathbb{E}[x_1 - x_0 \mid x_t = x]$. The pair $(x_1 - x_0, x_t)$ is jointly Gaussian, with $\mathbb{E}[x_1 - x_0] = \mu$, $\mathbb{E}[x_t] = t\mu$, $\mathrm{Cov}(x_1 - x_0, x_t) = t\Sigma - (1-t)I$ and $\mathrm{Cov}(x_t) = t^2\Sigma + (1-t)^2 I$. The conditional mean of a Gaussian vector is the linear regression. □

This is elementary and certainly known. Two consequences matter here. First, for independent standard Gaussian coordinates ($\mu = 0$, $\Sigma = I$) the field is $v = (2t-1)x/(2t^2 - 2t + 1)$ and the minimum of the loss Equation 3.3 per coordinate, averaged over t, is $\int_0^1 \mathrm{d}t/(2t^2 - 2t + 1) = \pi/2 \approx 1.571$: a model whose loss on normalised data is above $\pi/2$ does worse than this trivial field. Second, the network of Section 3.3 maps the latent to a hidden space of width 512 and back, so its Jacobian with respect to x has rank at most 512, whereas the linear part of Equation 3.4 is a full-rank, time-dependent matrix in a latent of 5320 dimensions. That the plain network cannot represent this linear part is exact; that this is the cause of its failure is a **conjecture**, tested by a width scan in the section Results (hypothesis H21). The remedy that does not rely on the conjecture is to give the model the term explicitly (W9) or exactly (W10, W11); Figure 3.4 summarises the argument.

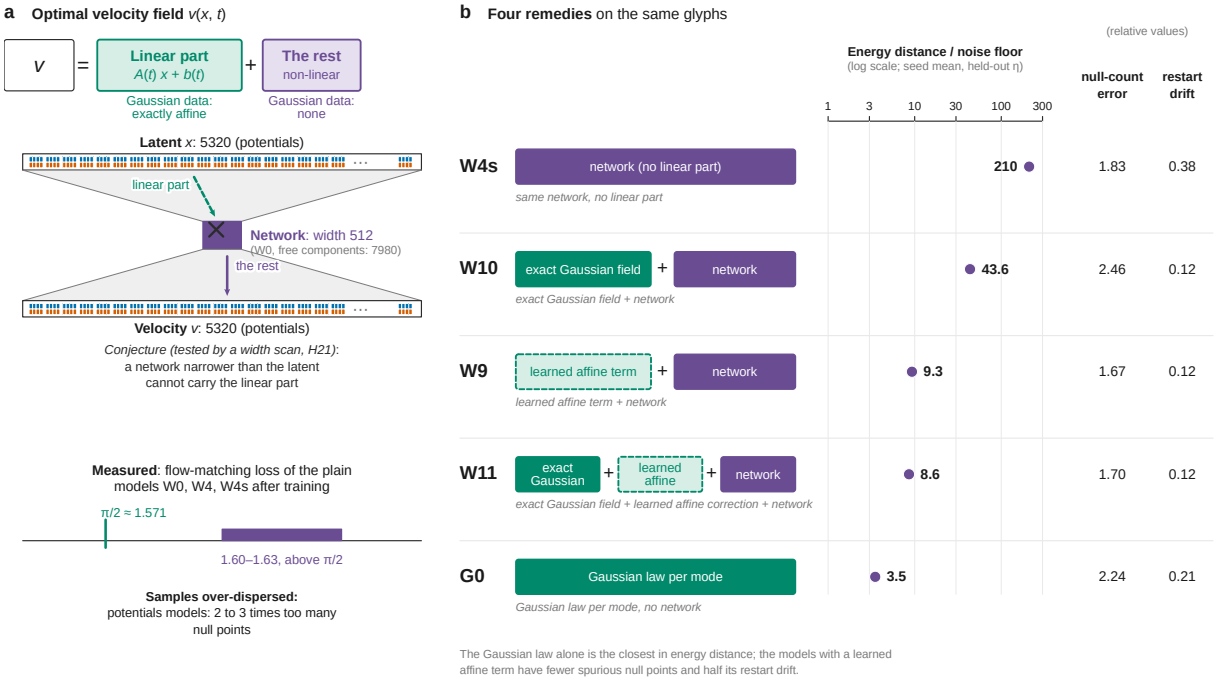


Figure 3.4: Why an explicit linear part. (a) The optimal velocity field of a Gaussian law is affine in x, with a full-rank matrix that depends on time; the plain networks end above the loss of the trivial Gaussian field; the statement on the width is a conjecture. (b) The four remedies and the same network without them: energy distance over the noise floor, null-count error and restart drift, seed means over five seeds on the held-out η (measured; `experiments/results/e4_mhd3d_s<k>.json`, values of Table 6.1). Schematic; source in `figures/linear-part/`.

3.6 Metrics

All metrics compare an ensemble of generated snapshots at one η with the reference ensemble at the same η (Figure 3.5).

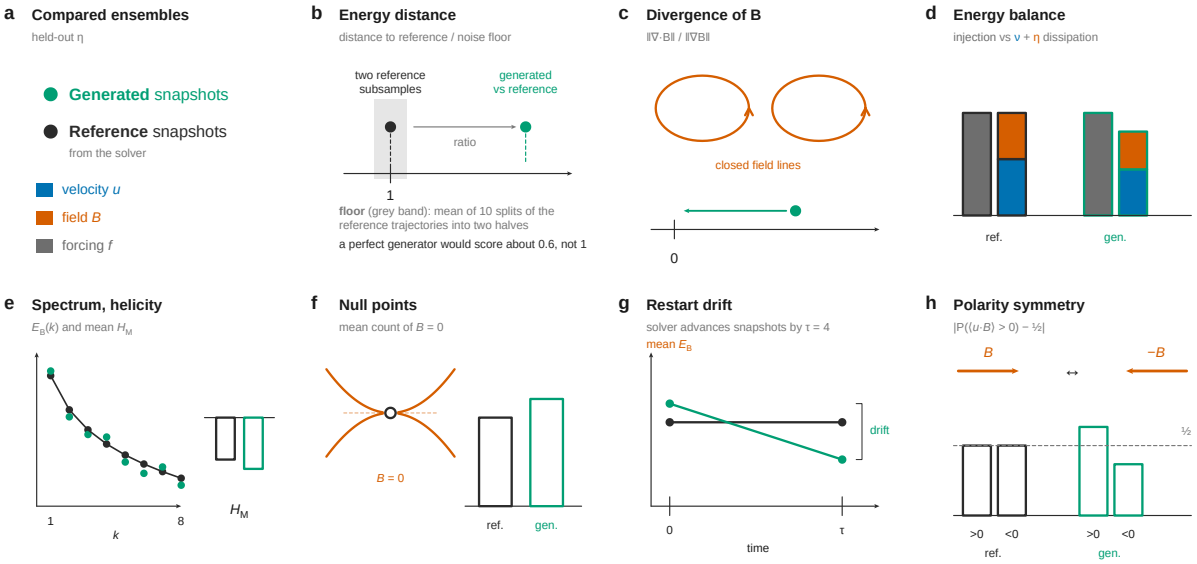


Figure 3.5: The metrics of E4: energy distance over its noise floor, divergence of B, energy balance, spectra and magnetic helicity, magnetic null points, restart drift, polarity. Schematic; source in `figures/metrics/`.

- **Energy-distance ratio.** Each snapshot is reduced to 21 physical features (logarithms of the shell spectra of u and B for $k = 1, \dots, 8$, relative kinetic, magnetic and cross helicities, logarithms of the two energies), standardised with the reference. The energy distance (Székely and Rizzo 2013) between 512 generated and 512 reference snapshots is divided by a **noise floor**: the same distance between two reference subsamples of 512 snapshots taken from disjoint halves of the 24 trajectories (mean of 10 splits). The ratio is the number of floors separating the generated law from the reference. A perfect generator would not give 1 but less, because its samples are independent whereas the reference trajectories are correlated in time (an estimate from the two kinds of floors of Figure 4.5 gives about 0.6; it was not measured with a generator); the floor is a scale, not a bound, and it is itself uncertain: its standard deviation over the splits is 26 to 99 % of its mean.
- **Divergence.** $\|\nabla \cdot B\|/\|\nabla B\|$ in the spectral norm, largest value over the ensemble.
- **Energy balance.** $|\mathbb{E}[\langle f \cdot u\rangle - D_\nu - D_\eta]|/\mathbb{E}[D_\nu + D_\eta]$ with the two dissipations $D_\nu = \nu\langle|\omega|^2\rangle$ and $D_\eta = \eta\langle|j|^2\rangle$.
- **Magnetic energy balance.** $|\mathbb{E}[P - D_\eta]|/(|\mathbb{E}[P]| + \mathbb{E}[D_\eta])$ with the production $P = \langle j \cdot (u \times B)\rangle$: zero when the dynamo term balances the Ohmic dissipation, one when there is no production at all.
- **Spectrum and helicity.** Relative L^2 error of the mean shell spectrum $E_B(k)$; relative error of the mean magnetic helicity.
- **Null points.** Relative error of the mean number of magnetic null points per snapshot. The field is evaluated on a 48^3 grid, and the nulls of its trilinear interpolant are found by the method of Haynes and Parnell (2007): cells where the three components change sign, then Newton iterations from nine starting points. On a periodic box the numbers of nulls of the two signs of $\det \nabla B$ must be equal (Poincaré–Hopf), which is used as a test of the implementation.
- **Restart drift.** The generated snapshots are used as initial conditions of the solver and advanced over $\tau = 4$ time units; the metric is the relative change of the ensemble-mean magnetic energy (and, reported with it, of the kinetic energy). It is zero for samples of the true stationary law, and it is the only metric here that tests the generated law against the dynamics rather than against statistics chosen in advance.

- **Polarity.** $|P(\langle u \cdot B \rangle > 0) - \frac{1}{2}|$, zero for a law invariant under $B \to -B$.

4 Experiments

4.1 Reference data

For each of the 11 values of η (logarithmically spaced from 0.08 to 0.03), 24 independent trajectories are started from a small random magnetic seed and a perturbed ABC flow, integrated over 200 time units to let the dynamo saturate, then sampled every 2 time units over 200 time units: 2400 snapshots of (u, B) per value (`experiments/e4_data.py`). Table 4.1 gives the statistics of the saturated state, all **measured**: the kinetic energy falls and the magnetic energy rises as η decreases; the magnetic helicity is negative and its magnitude grows; about 40 % of the magnetic energy sits in the largest scale $k = 1$; the number of magnetic null points per snapshot more than doubles across the range (Figure 4.1).

Table 4.1: Reference data of E4 (measured; `experiments/results/e4_data.json` and `e4_reference.json`). E_u, E_B: kinetic and magnetic energies; $H_M = \langle A \cdot B \rangle$; null points per snapshot (mean $\pm$ standard error over 400 snapshots); $k_\lambda = (\langle |j|^2 \rangle / \langle |B|^2 \rangle)^{1/2}$; energy residual: mean of injection minus dissipation, over the mean dissipation; floor: energy distance between two halves of the reference trajectories.

η	Rm	Split	E_u	E_B	H_M	$E_B(k{=}1)/E_B$	Null points	k_λ	Energy residual	Floor of the energy distance
0.0800	6.2	training	0.491	0.0783	-0.0433	0.436	21.1 $\pm$ 0.47	2.5	-3.0×10^{-5}	0.117
0.0725	6.9	held-out test	0.44	0.0967	-0.056	0.434	21.4 $\pm$ 0.53	2.53	4.4×10^{-6}	0.0703
0.0658	7.6	training	0.402	0.111	-0.0646	0.426	21.9 $\pm$ 0.49	2.57	-4.1×10^{-4}	0.0363
0.0596	8.4	validation	0.373	0.121	-0.0719	0.422	24.2 $\pm$ 0.56	2.61	1.3×10^{-4}	0.0425
0.0540	9.3	training	0.345	0.131	-0.0778	0.413	24 $\pm$ 0.62	2.65	2.2×10^{-4}	0.037
0.0490	10.2	held-out test	0.321	0.14	-0.0841	0.408	27.6 $\pm$ 0.65	2.69	-4.6×10^{-5}	0.0394

η	Rm	Split	E_u	E_B	H_M	$E_B(k{=}1)/E_B$	Null points	k_λ	Energy residual	Floor of the energy distance
0.0444	11.3	training	0.309	0.145	-0.0849	0.396	31.7 ± 0.76	2.74	-2.7×10^{-4}	0.0409
0.0403	12.4	validation	0.288	0.153	-0.0904	0.389	32.5 ± 0.77	2.79	-2.1×10^{-4}	0.0514
0.0365	13.7	training	0.276	0.16	-0.0947	0.384	38.1 ± 0.98	2.84	6.6×10^{-6}	0.0479
0.0331	15.1	held-out test	0.265	0.165	-0.0962	0.371	44.1 ± 1.1	2.9	1.7×10^{-4}	0.0733
0.0300	16.7	training	0.253	0.172	-0.0968	0.364	52 ± 1.2	2.95	2.3×10^{-4}	0.04

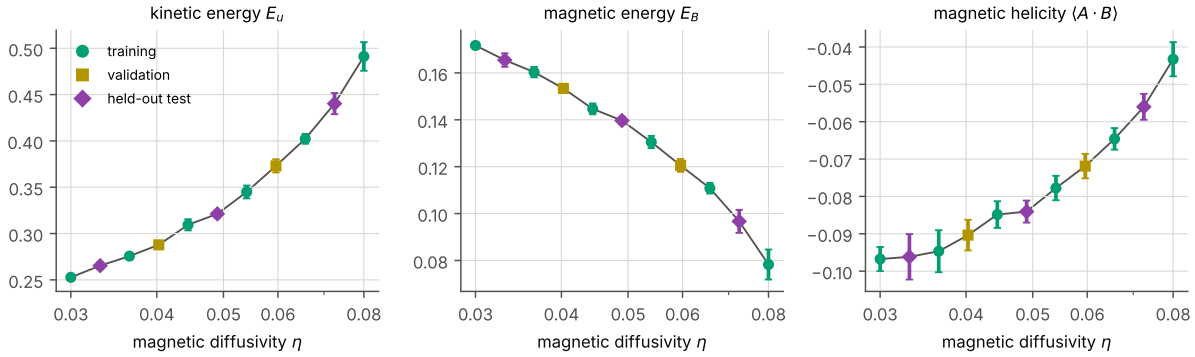


Figure 4.1: Reference data: kinetic energy, magnetic energy and magnetic helicity against η (mean over 24 trajectories; bars: ± 2 standard errors), with the training, validation and held-out values. Data plot from `experiments/results/e4_data.json` (`experiments/e4_plots.py`).

Two properties of the reference were checked before any model was trained.

The stationarity identities hold on the data. The energy residual of Table 4.1 is at most a few 10^{-4} of the dissipation at every η. The mean current helicity $\langle j \cdot B \rangle$ is zero within its standard error: its shell spectrum changes sign, negative at $k \leq 2$ and positive for $k \geq 3$ (Figure 4.2). A hypothesis registered beforehand (H11) predicted a simpler picture, a magnetic helicity spectrum negative at $k = 1$ only and positive for every $k \geq 2$; it was **refuted** by the data, the shell $k = 2$ being negative as well.

The data are converged for the purpose. The same statistics were recomputed at the three held-out η with half the time step and on a 24^3 grid, and at one η on a 32^3 grid (Figure 4.3): energies and helicity agree with those of the 16^3 data within one to two standard errors. The flow at

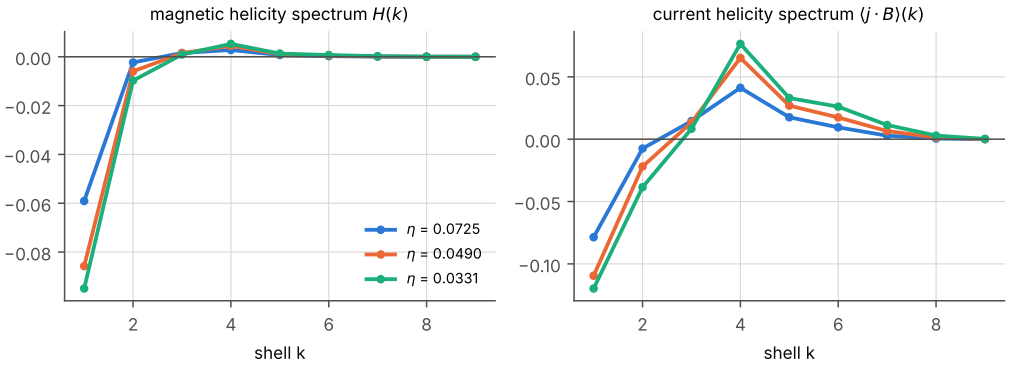


Figure 4.2: Reference data: shell spectra of the magnetic helicity and of the current helicity $\langle j \cdot B \rangle$ at the three held-out η. The mean of $\langle j \cdot B \rangle$ is zero: the negative shells $k \leq 2$ cancel the positive shells $k \geq 3$. Data plot from `experiments/results/e4_reference.json`.

$Re = 10$ is smooth; the small grid is a choice of scale for the comparison, not an under-resolved simulation.

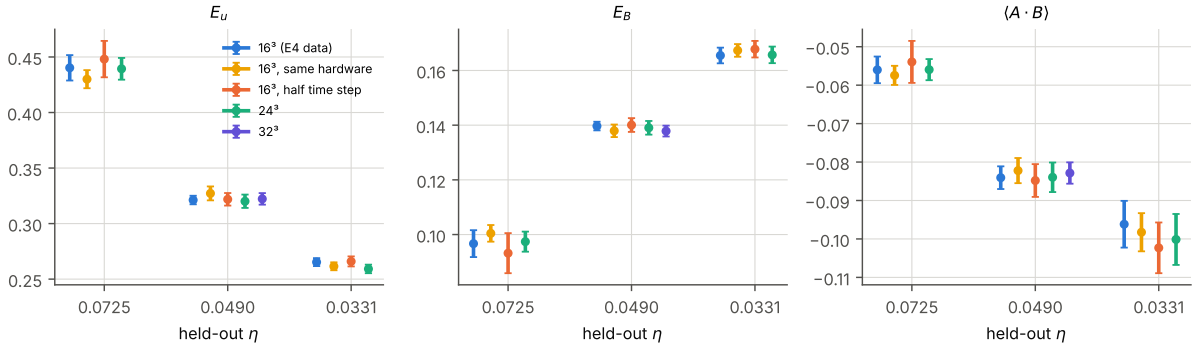


Figure 4.3: Convergence of the reference data: energies and magnetic helicity at the three held-out η for the 16^3 grid of E4, the same grid recomputed on the hardware of the check, half its time step, 24^3 and 32^3 (one η); bars: ± 2 standard errors over 24 trajectories. Data plot from `experiments/results/replications/e4_convergence/`.

4.2 Protocol

Split of the parameter values. The 11 values alternate (Figure 4.4): six training values (the two ends and every second value), two validation values and three held-out test values. The test values are used neither to train nor to choose a model. Every number quoted "on the held-out η" is the mean over the three test values.

Seeds. Each variant is trained with five seeds. A comparison between two variants is the paired difference per seed, quoted as mean $\pm$ standard error over the five seeds (rule 4 of `AGENTS.md`).

Noise floor. The floor of the energy distance (Section 3.6) is computed per η from disjoint halves of the trajectories and not from a random split of the snapshots: snapshots of one trajectory are correlated in time, and a random split gives a floor three to six times lower that no independent run of the solver would reach (Figure 4.5).

Training and sampling. 4000 iterations of Adam, batch 128, learning rate 5×10^{-4}, exponential moving average of the weights; 512 samples per validation and test value, 256 per training value.

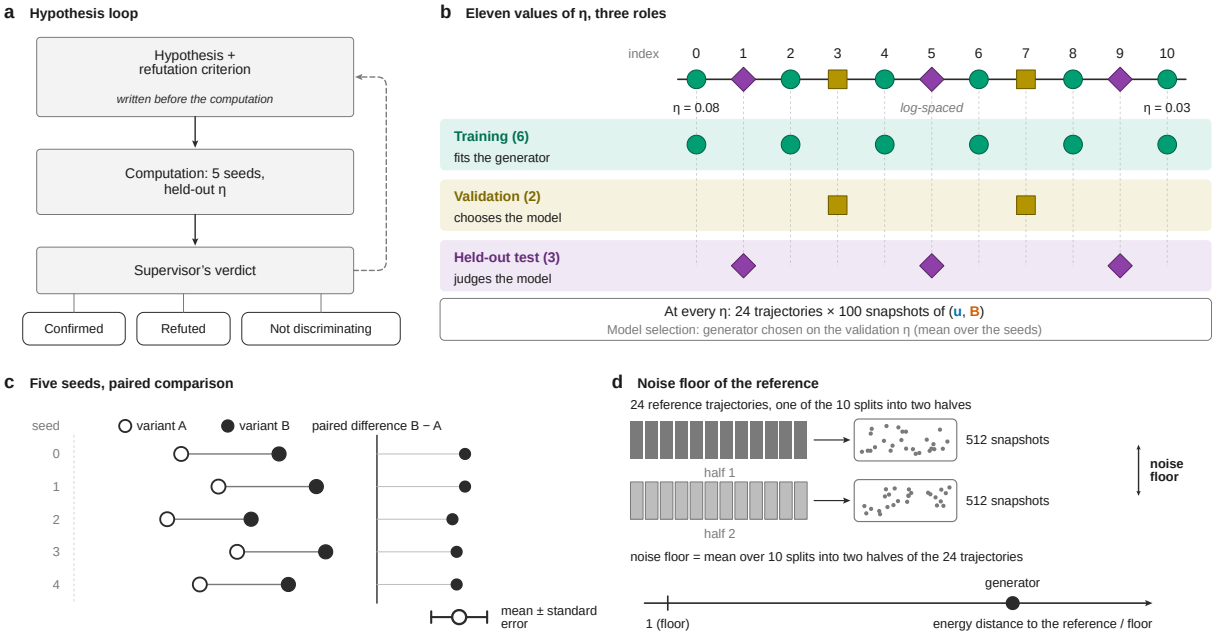


Figure 4.4: The protocol of E4: values of η for training, validation and held-out test; five model seeds; noise floor of the energy distance from two halves of the reference trajectories; hypotheses registered before the computation and reviewed afterwards. Schematic; source in `figures/protocol/`.

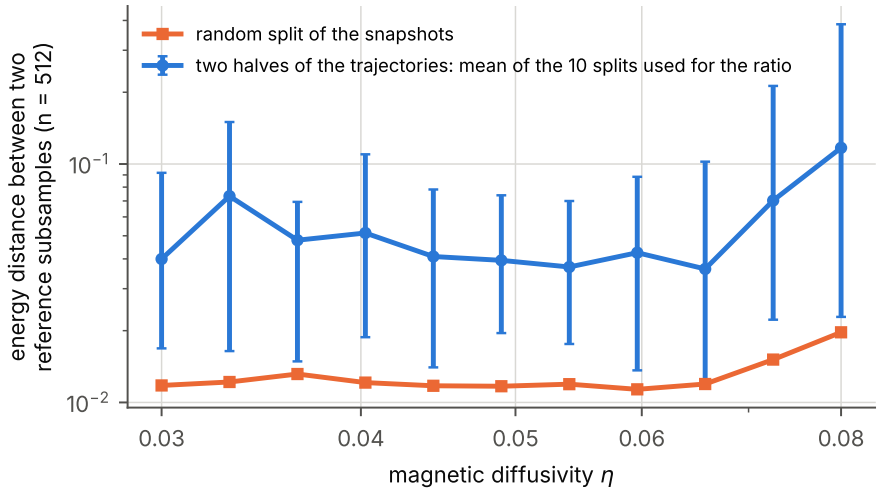


Figure 4.5: Noise floors of the energy distance on the reference data: between two subsamples from disjoint halves of the 24 trajectories (mean of the 10 splits that define the floor; bars: smallest and largest of 20 splits) and between two random subsamples. Data plot from `experiments/results/e4_floor.json`.

The “long” variants use 8 times the iterations. The null points are counted on 200 samples per value and the restart test uses 128 samples.

Register and review. Each hypothesis is written in `research/hypotheses.json` with its refutation criterion before the corresponding run, and the commit of the registration precedes the commit of the results. The `encadrant` role then recomputes the comparison from the result files and gives one verdict per criterion; its reports are in `research/reviews/`. The selection of the generator that is measured against the objectives of the race uses the validation values only.

4.3 Compute

The session runs in a cloud container with four CPU cores and no GPU. The data, the references and the first two seeds were computed there; the five-seed runs were made on rented GPU pods through `tools/runpod_job.py`, which clones the pushed commit on the pod, runs the command under a time limit, brings the outputs back, deletes the pod and writes the cost in `research/compute-ledger.jsonl` (Figure 4.6). A training of 4000 iterations takes 30 s to 2 minutes on the GPU, depending on the variant and on how many seeds share the card, against about 15 minutes on one CPU core; the evaluation (null points, restart test) dominates the run time. The costs are in the section Reproducibility.

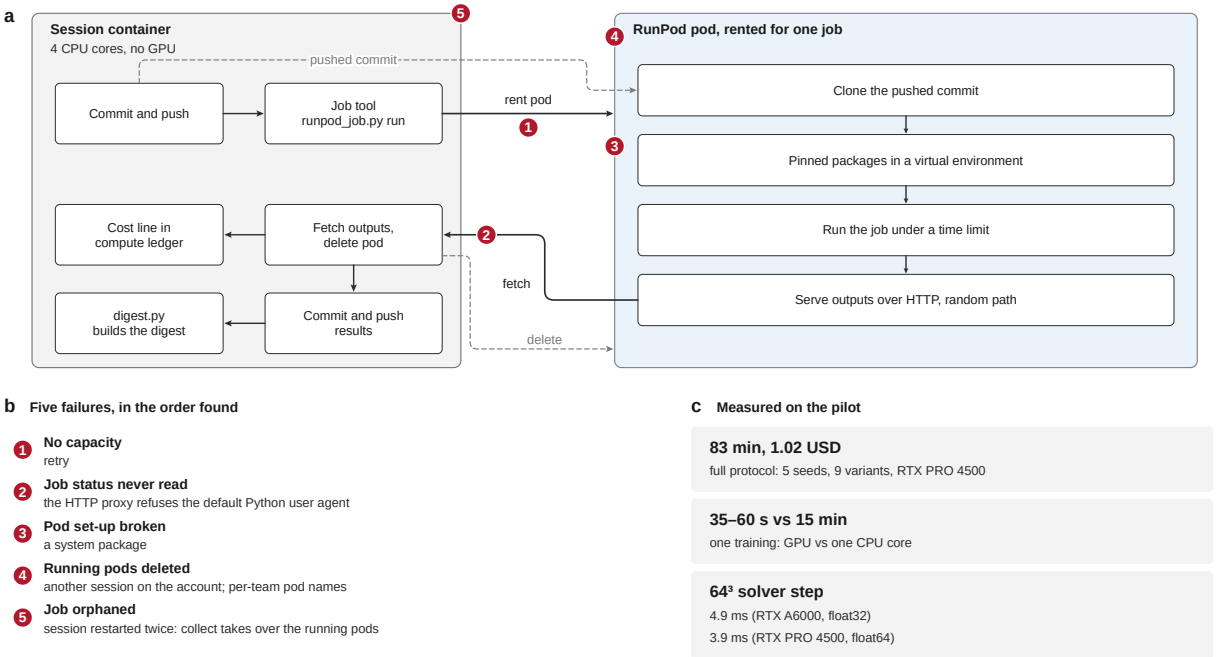


Figure 4.6: The compute path of the pilot: session, GPU pod, results, ledger, and the failures met and repaired on the way. Schematic; source in `figures/compute-path/`.

The workflow of the experiment and the organisation of the agents are drawn as two explorable diagrams, experiment-workflow.html and agent-organisation.html; their static exports are Figure 4.7 and Figure 4.8.

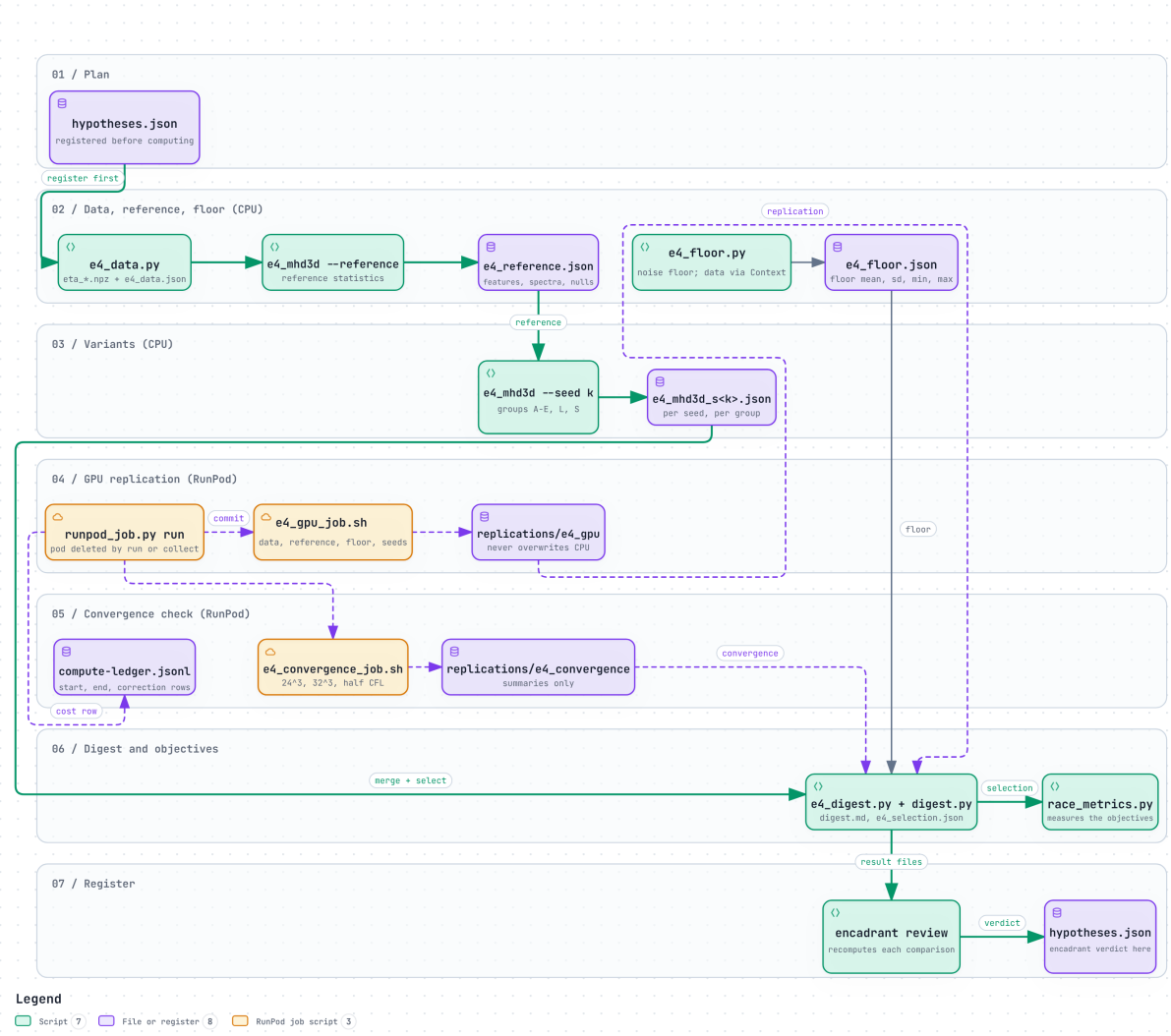


Figure 4.7: Workflow of experiment E4: plan, data, reference and floors, variants, GPU runs, digest, objectives and register. Static export of the Archify diagram `diagrams/experiment-workflow.html`.

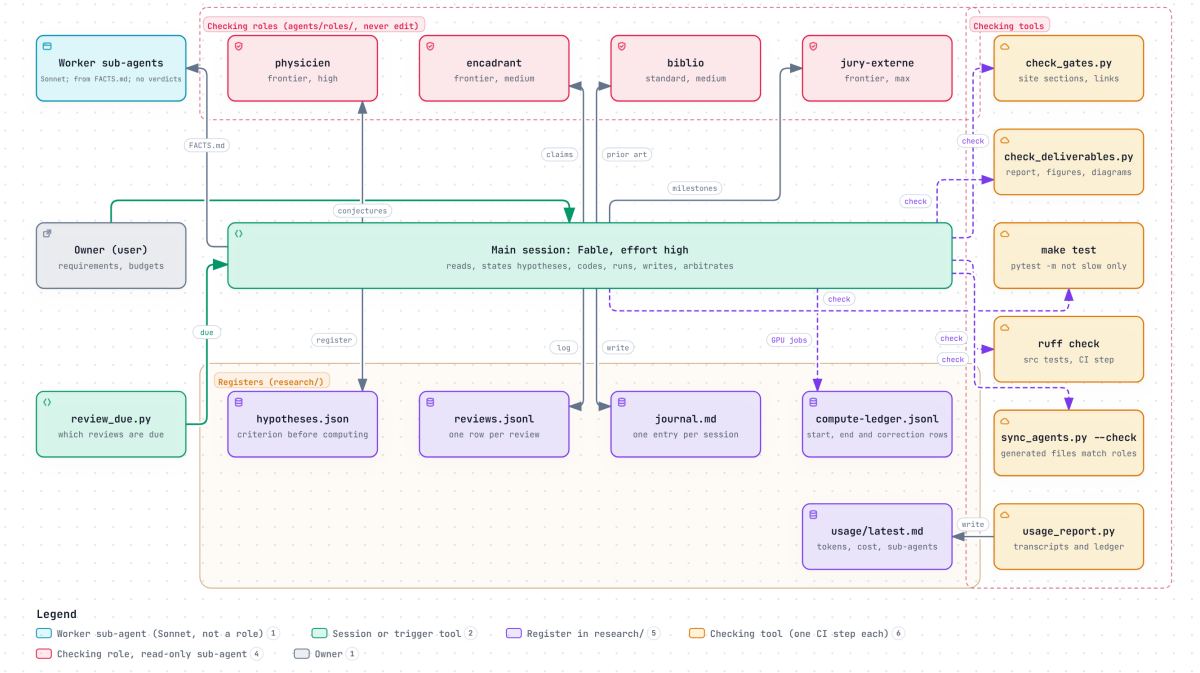


Figure 4.8: Organisation of the team: the main session, the four roles with their model tier and reasoning level, the reviews named by `tools/review_due.py`, the registers and the checking tools. Static export of the Archify diagram `diagrams/agent-organisation.html`.

5 Results

Everything in this section is **measured**, on five model seeds unless a table says otherwise; the numbers are in Table 6.1 of the section Comparative study and in the tables below.

5.1 The registered protocol: every first-generation model is far from the reference

The first protocol, registered before any training, compared the baseline W0, the mechanisms W1 to W5, the diffusion model W6 and the sampling-time corrections W7 and W8, all with the same residual network of width 512 (Figure 5.1).

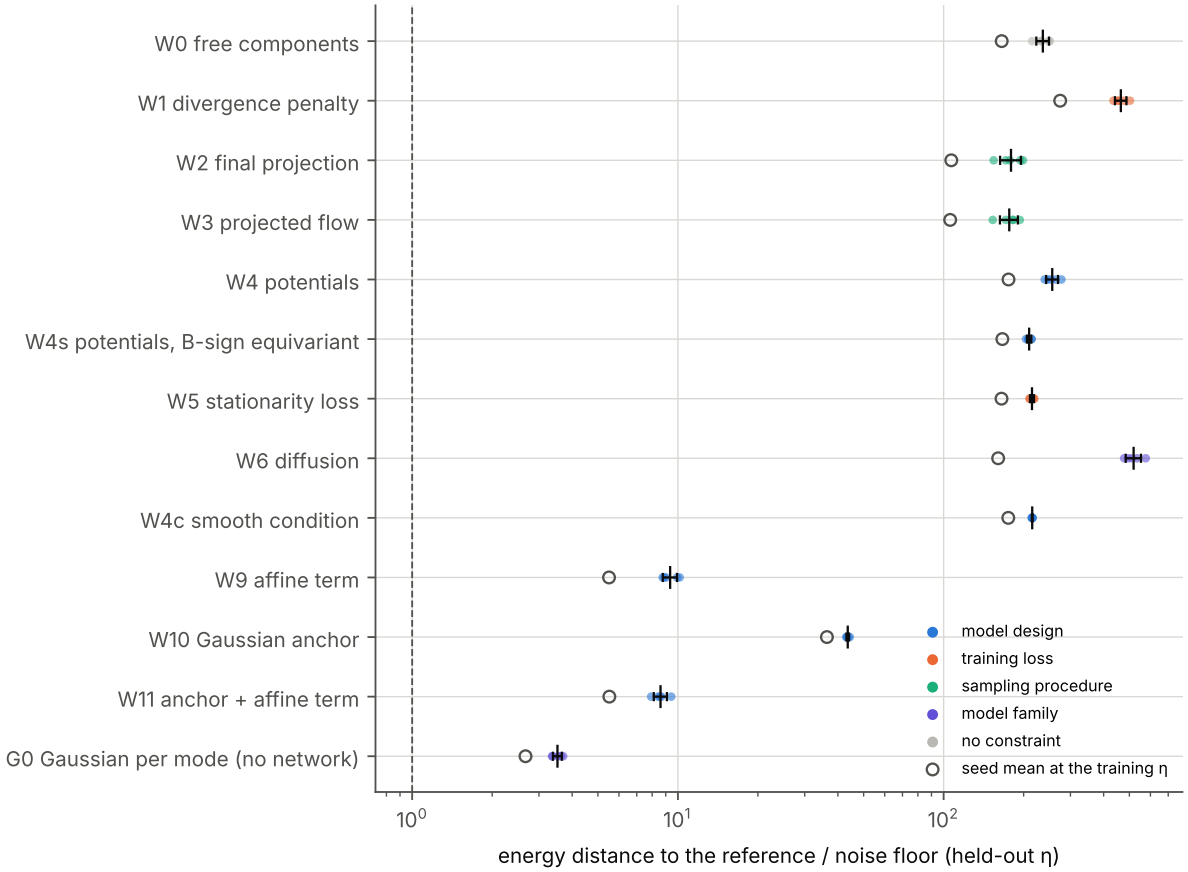


Figure 5.1: Energy distance to the reference divided by the noise floor on the held-out η, per variant: one dot per seed, bar: seed mean ± 2 standard errors (drawn with at least five seeds), open circle: seed mean at the training η; dashed line: the floor. Data plot from `experiments/results/e4_mhd3d_s<k>.json`.

The outcome is negative for all of them: the generated laws are between about one hundred and five hundred floors away from the reference on the held-out values, and not much closer on the

training values. Within this failure the mechanisms can still be told apart, and the registered hypotheses were decided as follows (verdicts of the `encadrant`, Table 6.2).

- **Divergence.** The potentials (W4), the final projection (W2) and the projected flow (W3) bring the relative divergence of B to round-off, about 10^{-8}; the free model stays at order one and the penalty (W1) only halves it (H5 confirmed). The divergence of the free model does not change when the number of integration steps is multiplied by four, from 50 to 200 (H14 confirmed): it is an error of the learned field, not of the integrator, as Proposition 2 of the theory brief predicts for a field that lacks the contraction towards the constraint set.
- **Potentials against free components.** Generating potentials does not bring the law closer to the reference than generating free components: the paired difference has the wrong sign in all five seeds (H4 refuted). The projected flow is not better than a projection of the final samples (H6 refuted); the two are the best of this group.
- **The penalty costs accuracy.** The model trained with the divergence penalty is about twice as far from the reference as the baseline. The penalty is unbiased at the optimum, but the network is far from the optimum, and there the penalty competes with the regression.
- **Stationarity loss.** Fine-tuning with the law-level identities lowers the residual of the mean energy balance by about a third against the control that receives the same fine-tuning without the term (H7 confirmed), at parameter values where no data were used. This is the effect on the energy identity only. The residual of the magnetic-helicity identity, the second term of the loss, does not move (-0.004 ± 0.009), and the loss has side effects that the registered criterion did not look at: against the control, the residual of the magnetic energy balance doubles (0.25 to 0.52), the error on the null count rises by 0.56, the error on the magnetic spectrum falls by 0.56, and the distance to the reference law is unchanged. The comparison between imposing the identities in the loss and by tilting was dropped as not discriminating (H8): the tilted ensemble is one weighted sample, and the registered criterion could be met by its sampling noise.
- **Which identity constrains more.** The registered conjecture that the magnetic-helicity identity is violated more than the energy identity was refuted (H12): on the samples of the equivariant model the normalised residual of $\mathbb{E}[\langle j \cdot B \rangle] = 0$ is about half that of the energy balance.
- **Equivariance.** The polarity asymmetry of the non-equivariant model is already at the sampling noise of 512 samples; the equivariant model cannot lower it (H9 refuted on its first clause). The symmetry is exact by construction in W4s, but in this system it is not what the unconstrained model gets wrong.
- **Flow matching against diffusion.** At the same number of network evaluations, flow matching is about twice closer to the reference than the diffusion model on the held-out η (H10 confirmed as registered). On the training η the order is reversed: the diffusion model is the closer one (160 against 175 floors, in every seed). The advantage of flow matching is in the interpolation in η (Figure 5.2), not in the family at a fixed parameter value.
- **Per-sample rescaling.** W8 satisfies the energy balance of each sample exactly, and has the lowest distance of this group; its law is still about a hundred floors from the reference, and by construction the variance of $\mathrm{d}E/\mathrm{d}t$ of its samples is zero.

5.2 Diagnosis: the models do worse than a Gaussian field

Three measurements point to the same cause. The flow-matching loss of the plain flow-matching models (W0, W4, W4s and the control of W5; the other variants have another loss or another noise and are not comparable), evaluated on training samples, ends between 1.60 and 1.63,

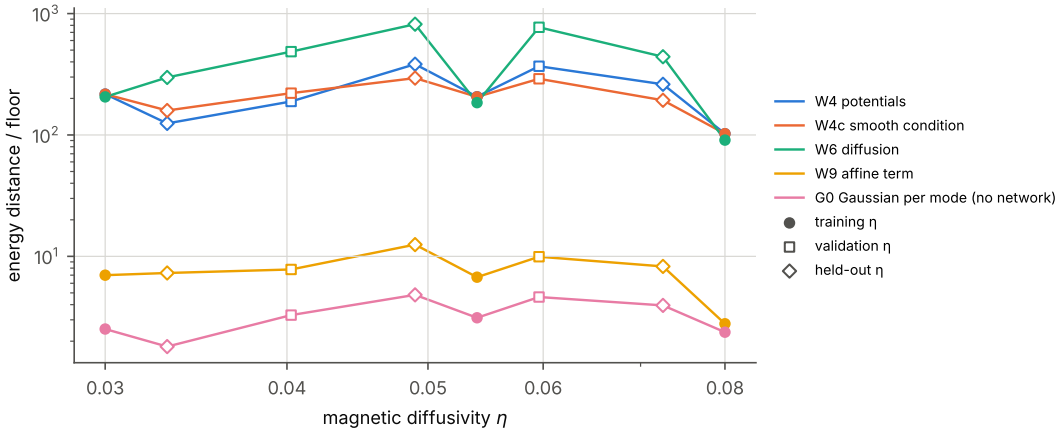


Figure 5.2: Energy-distance ratio against η, seed mean: at the training values (filled circles) and between them (open squares: validation, open diamonds: held-out test). Data plot from `experiments/results/e4_mhd3d_s<k>.json`.

above $\pi/2 \approx 1.571$, the loss of the exact field for independent standard Gaussian coordinates (Section 3.5): after training, these networks fit the normalised data worse than a field that ignores every correlation. The samples of the potentials models are over-dispersed: about two to three times too many magnetic null points (the penalty model W1 and the diffusion model W6 err the other way, with too few) and a fluctuation of the energy budget several times that of the reference. The losses quoted here come from the training logs of the first runs (seeds 0 to 2); the result files store the final loss only for the later variants. And the error is of the same order on the training values as on the held-out ones, so it is not an error of interpolation in η.

Two explanations were registered and tested separately.

The embedding of the condition (H15). The Fourier embedding of the condition used frequencies of up to 8 cycles over the range of $\log \eta$, above what six training values can resolve. A model with a smooth embedding (W4c, frequencies up to half a cycle) was compared with W4.

The linear part of the field (H16, H17). The Gaussian baseline G0 has no network at all; the affine term of W9 gives the network of W4s the linear part of the field explicitly.

The measurements (five seeds, Table 6.1; paired differences in the digest; verdicts of the `encadrant` in Table 6.2) separate the two explanations.

- *The embedding matters for interpolation only.* With the smooth embedding the ratio on the held-out η falls from 256 to 215 (paired difference 40.9 ± 6.8, positive in the five seeds), and on the training η nothing changes (175 for both). The registered criterion asked for a factor of two: H15 is **refuted**, the embedding is a real but minor part of the error.
- *A Gaussian law without a network is fifty times closer.* G0 is at 3.5 floors on the held-out η, against 177 for the best trained model of the first protocol, the projected flow W3 (H16 **confirmed**).
- *The affine term is what was missing.* The same equivariant network with the explicit mode-wise affine term (W9) goes from 210 to 9.3 floors, a factor of 22, in every seed (H17 **confirmed**), and its flow-matching loss on training samples falls from 1.6 to 1.18, below $\pi/2$.

5.3 Giving the field its linear part

Four ways of giving the velocity field its linear part were compared (Table 6.1, Figure 5.1, Figure 5.3, Figure 5.4).

- **Sampling the Gaussian law itself (G0)** is the closest to the reference in energy distance (3.5 floors), with a residual of the total energy balance of 0.0015 and spectra within a few per cent. It is also the clearest illustration of what a distance built on second-order features does not see. Its magnetic energy balance is violated completely (residual 0.998): the law has the right Ohmic dissipation and no production, as the remark of Section 3.2 predicts for independent Fourier modes. It has more than three times the reference number of magnetic null points (relative error 2.24). And when the solver restarts from its snapshots the magnetic energy drifts by 21 % in four time units, against less than 1 % for reference snapshots. The Gaussian law has the spectra of a dynamo and none of its dynamics.
- **Learning the affine term next to the network (W9)** gives 9.3 floors, and **anchoring the field on the exact Gaussian field with a learned affine correction (W11)** gives 8.6; the two are indistinguishable within the seed noise on every metric. Both are about 2.5 times farther from the reference than G0 in energy distance, with a residual of the total energy balance of 0.05 to 0.06, a magnetic spectrum within 15 % and a magnetic helicity within 20 %. But they restore what the Gaussian law lacks: the residual of the magnetic energy balance falls from 0.998 to 0.055 and 0.060, the error on the null count from 2.24 to 1.7, and the restart drift of the magnetic energy from 0.21 to 0.12, in every seed. The restart drift of the kinetic energy goes the other way (0.026 for G0, 0.053 for W9 and W11).
- **Anchoring on the exact Gaussian field with a plain network (W10)** is worse than the Gaussian law it starts from: 43.6 floors (H19 refuted). The registered explanation (H20) was that the network's own linear response spoils the second-order statistics that the anchor had right, and that a learnable affine correction would restore them. The correction does bring the model from 43.6 to 8.6 floors, but not below G0, and H20 was refuted on that clause; its two other clauses, on the restart drift and the null count, are met.

So on this system the flow-matching models that carry their linear part learn part of what a Gaussian law cannot hold, the third-order correlations that make the field a dynamo, and pay for it with a less accurate second-order structure. None of the generators reproduces the topology. The errors on the null count go both ways (too few nulls for the diffusion and penalty models, too many for all the others), and the models closest to the reference law have 2.7 times (W9, W11) to 3.2 times (G0) the reference number of null points.

5.4 Two Gaussian laws: what the covariance between modes holds

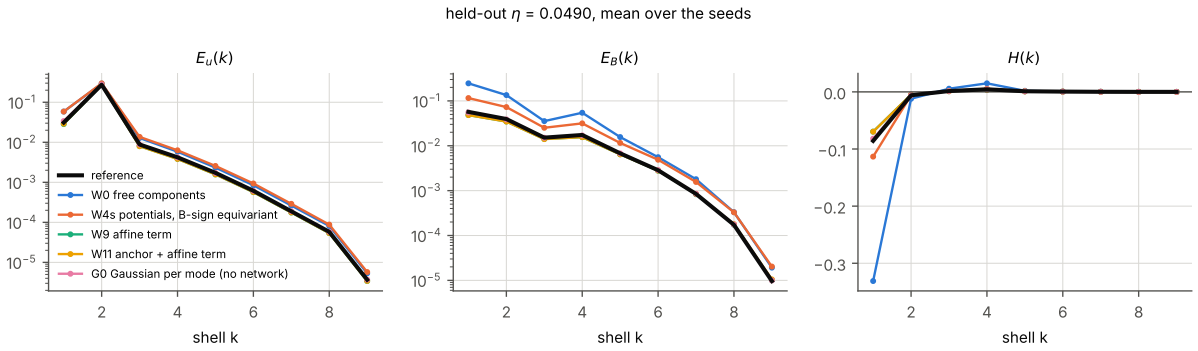


Figure 5.3: Shell spectra of the velocity and of the magnetic field and magnetic helicity spectrum at one held-out η: reference (black) and generated ensembles (mean over the seeds). Data plot from `experiments/results/e4_mhd3d_s<k>.json` and `e4_reference.json`.

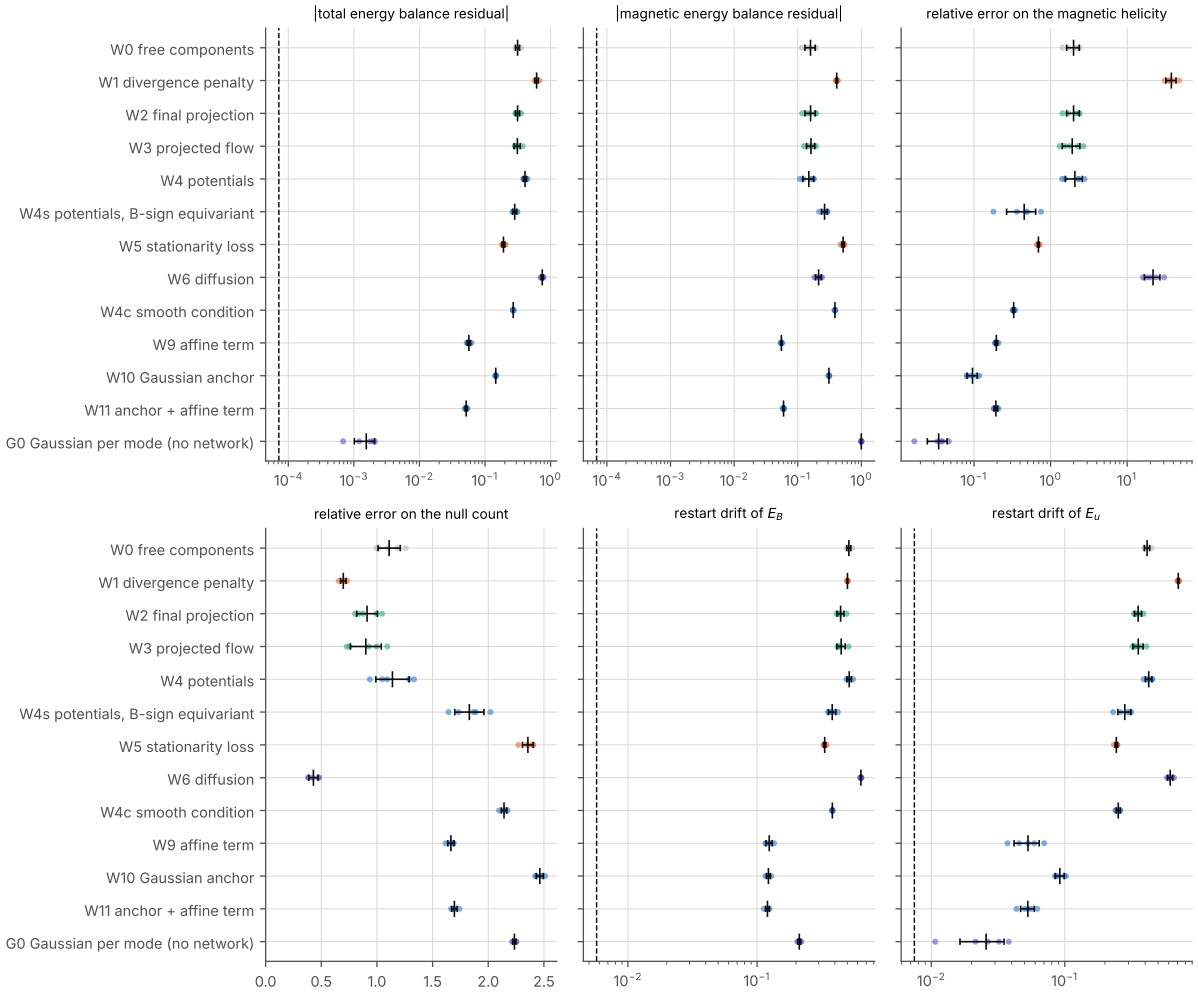


Figure 5.4: Physical metrics on the held-out η, one dot per seed and seed mean ± 2 standard errors: residuals of the mean total and magnetic energy balances, errors on the mean magnetic helicity and on the number of magnetic null points, drifts of the magnetic and kinetic energies when the solver restarts from the generated snapshots. Dashed lines: the same quantity on reference snapshots. Data plot from `experiments/results/e4_mhd3d_s<k>.json`.

: Gaussian laws and mechanisms on the model with the affine term (measured; `experiments/results/replications/e4_mech`, five seeds, held-out η; the models G0, W9 and W11 are retrained in this job). G1: Gaussian law with the full covariance of the latent. W12: W9 fine-tuned with the stationarity loss, and its control. W13: exponential tilting of the second half of the control samples, to compare with the untilted half. Columns and notation as in Table 6.1. {#tbl-mech}

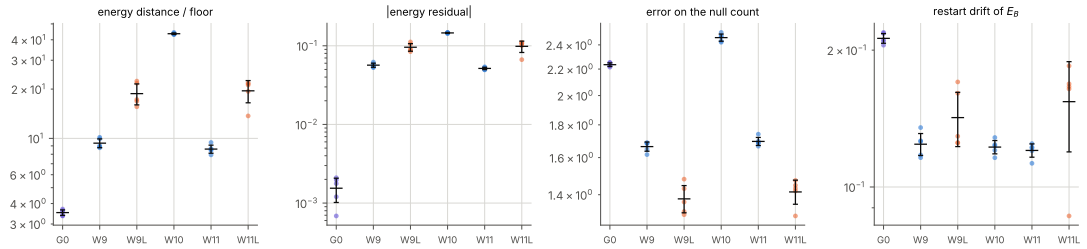


Figure 5.5: Restart test: signed relative change of the ensemble-mean magnetic and kinetic energies when the solver restarts from generated snapshots, at four times, mean over the held-out η and the seeds (bars: ± 2 standard errors). Data plot from `experiments/results/replications/e4_mech/e4_mhd3d_s<k>.json`.

5.5 Loss and sampling mechanisms on a model close to the reference

5.6 Longer trainings and wider networks

Would the models with an affine term overtake the Gaussian law with more training? The two best models were retrained with eight times the iterations (W9L, W11L; Table 5.1, Figure 5.6), in a separate job whose base models reproduce those of the main run to the last digit.

The answer is no, and the result is a negative one. The flow-matching loss on training samples goes down (from 1.18 to 1.08), but the generated law moves away from the reference: 18.8 and 19.5 floors on the held-out η against 9.3 and 8.6, and 12 against 5.5 on the training η themselves. The longer training improves the topology (error on the null count 1.4 against 1.7) and keeps the magnetic energy balance (residual 0.05), while the total energy balance and the restart drifts get somewhat worse. Why a lower training loss gives a worse law was not established; a decrease of the loss that sharpens the velocity field beyond what 50 integration steps resolve is one **conjecture**, memorisation of the 14 400 training snapshots another.

The hypothesis registered for this run (H18) was narrower: that with the affine term and a long training, flow matching holds what a Gaussian law cannot, a lower restart drift of the magnetic energy and a lower error on the null count than G0. Both paired differences are positive in the five seeds (0.070 ± 0.011 and 0.86 ± 0.03).

Table 5.1: Long trainings (measured; `experiments/results/replications/e4_final_long`, five seeds, held-out η unless said otherwise): Gaussian baseline, models with the affine term or the anchor at the base budget and with 8 times the iterations. Columns and notation as in Table 6.1; FM loss: flow-matching loss on a training sample ($\pi/2 = 1.571$ for the exact field of independent standard Gaussians).

Variant	Acts through	ED / floor, training η	ED / floor	div B	energy residual	magnetic balance	spectrum of B	magnetic helicity	null points	restart drift E_B	restart drift E_u	polarity	FM loss	Seeds
G0 Gaussian per mode (no network)	family	2.67(8)	3.52(7)	1.5×10^{-8}	0.0015(3)	0.9981(9)	0.024(3)	0.035(5)	2.235(7)	0.212(3)	0.026(5)	0.018(3)	–	5
W9 mode-wise affine term	design	5.51(9)	9.3(3)	1.5×10^{-8}	0.0567(17)	0.0553(6)	0.145(3)	0.196(4)	1.665(14)	0.124(3)	0.053(6)	0.019(4)	1.1760(15)	5
W9L affine term, 8× iterations	design	11.91(15)	18.8(14)	1.5×10^{-8}	0.096(5)	0.049(4)	0.112(15)	0.12(2)	1.38(3)	0.142(10)	0.077(11)	0.0147(14)	1.075(2)	5
W10 anchored on the Gaussian field	design	36.37(10)	43.6(2)	1.5×10^{-8}	0.1458(9)	0.3092(17)	0.2081(19)	0.096(7)	2.464(16)	0.122(2)	0.092(4)	0.0135(9)	1.2727(18)	5
W11 anchor + affine term	design	5.52(13)	8.6(2)	1.5×10^{-8}	0.0515(9)	0.0598(6)	0.143(4)	0.193(5)	1.696(12)	0.120(2)	0.053(3)	0.0169(10)	1.1742(14)	5
W11L anchor + affine, 8× iterations	design	12.0(5)	19.5(15)	1.5×10^{-8}	0.098(8)	0.043(3)	0.128(14)	0.14(2)	1.41(3)	0.154(17)	0.081(13)	0.018(5)	1.075(2)	5

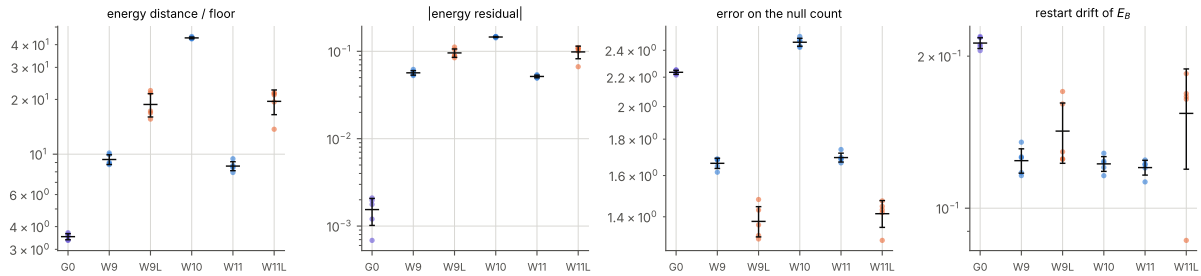


Figure 5.6: Gaussian baseline (violet), flow matching with the explicit affine term or the Gaussian anchor at the base budget (blue) and with 8 times the iterations (orange), on the held-out η: one dot per seed, seed mean ± 2 standard errors. Data plot from `experiments/results/replications/e4_final_long/`.

Table 5.2: Width scan (measured; `experiments/results/replications/e4_final_scan`): the equivariant potentials model with the smooth embedding and no affine term, for three widths and two training budgets, against the model with the affine term and the Gaussian baseline.

Variant	Acts through	ED / floor, training η	ED / floor	div B	energy residual	magnetic balance	spectrum of B	magnetic helicity	null points	restart drift E_B	restart drift E_u	polarity	FM loss	Seeds
S width 512	family	166.0(5)	205.6(5)	1.4×10^{-8}	0.2737(9)	0.328(2)	0.665(6)	0.186(7)	2.124(11)	0.3725(20)	0.268(3)	0.0165(11)	1.608(2)	5
S width 2048	family	25.3(2)	25.3(3)	1.5×10^{-8}	0.0130(6)	0.0960(13)	0.143(4)	0.278(6)	2.004(20)	0.072(4)	0.075(4)	0.0141(12)	1.2169(17)	5
S width 5632	family	76(5)	99(8)	1.5×10^{-8}	0.270(19)	0.028(3)	0.35(2)	0.37(2)	1.00(6)	0.46(5)	0.29(4)	0.0215(20)	0.769(3)	2
S width 512, 8× iterations	family	175.1(14)	215.8(16)	1.4×10^{-8}	0.2607(20)	0.398(8)	0.568(4)	0.34(2)	2.187(5)	0.364(3)	0.223(7)	0.013(6)	1.526(6)	2
W9 mode-wise affine term	design	5.48(11)	9.40(19)	1.5×10^{-8}	0.0572(14)	0.0541(5)	0.148(3)	0.200(3)	1.678(14)	0.127(3)	0.050(6)	0.018(3)	1.1770(13)	5
G0 Gaussian per mode (no network)	family	2.67(8)	3.52(7)	1.5×10^{-8}	0.0015(3)	0.9981(9)	0.024(3)	0.035(5)	2.235(7)	0.212(3)	0.026(5)	0.018(3)	–	5

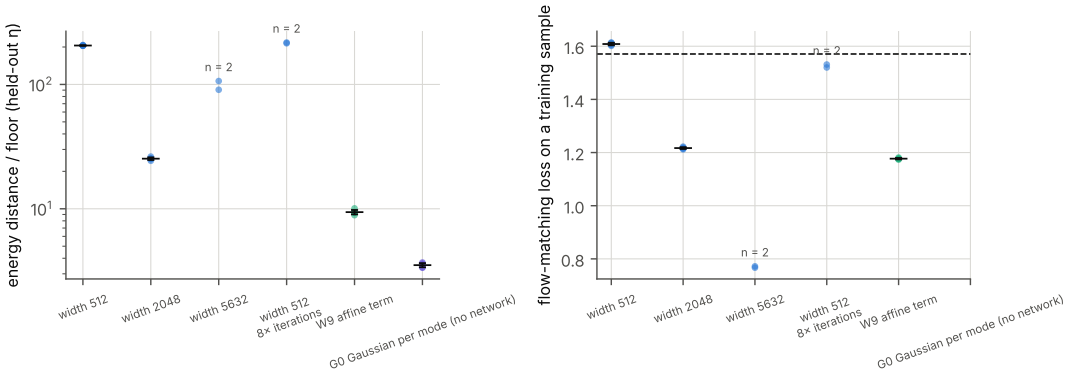


Figure 5.7: Width scan (hypothesis H21): energy-distance ratio and flow-matching loss for three widths and two budgets, against the model with the explicit affine term and the Gaussian baseline. Dashed line: $\pi/2$. Data plot from `experiments/results/replications/e4_final_scan/`.

5.7 Topology: null points follow the small scales

The number of magnetic null points is the most demanding of the metrics: no generator has it right. The penalty model and the diffusion model have too few (0.3 and 0.6 times the reference count); all the others have too many. A conjecture of the `physicien` role (H13) was that the count scales as the cube of the magnetic Taylor wavenumber k_λ, as it would for a Gaussian random field, so that the error on the count would be three times the relative error on k_λ. The fit over all generators, held-out values and seeds gives a slope close to five and not three (Figure 5.8): H13 was **refuted** on the slope. The correlation is strong, but it does not explain the excess: the fitted line does not pass through the origin, and a generator with exactly the reference k_λ would still have about 2.8 times the reference number of null points. The excess of nulls is a property of the generated fields beyond their small-scale energy.

5.8 Replication with other data

The five-seed runs made on pods of the same type reproduce one another to the last digits, because the data and model seeds are the same: they check the reproducibility of the pipeline, not the robustness of the conclusions. For that, the whole protocol (groups A to E, five seeds) was run once more on reference data regenerated with other seeds of the initial conditions (Table 5.3). The statistics of the two reference sets agree; the noise floors differ by up to a factor of 1.6 at one held-out value, which moves the ratios by 5 to 15 %.

Every conclusion of this section is unchanged in the replication: the first-generation models are between 110 and 490 floors from the reference, in the same order; the affine term brings the equivariant model from 191 to 8.2 floors (a factor of 23); the mode-wise Gaussian law is closer (2.9) and has no magnetic production; the anchored model without correction stays at 43 floors; the null-count errors and restart drifts keep their order.

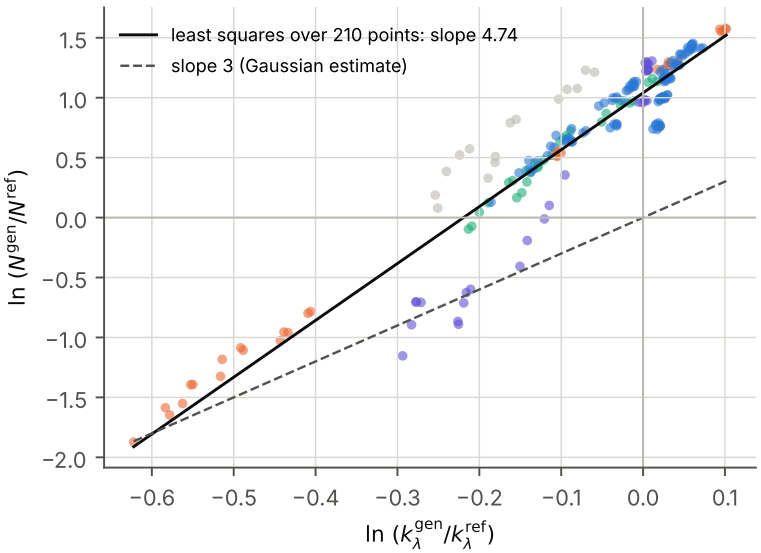


Figure 5.8: Number of magnetic null points against the magnetic Taylor wavenumber, generated over reference, one point per variant, held-out η and seed; the fit is the one of the digest (hypothesis H13). Data plot from `experiments/results/e4_mhd3d_s<k>.json`.

Table 5.3: Main run ("main") against a replication in which the reference data were regenerated with other seeds of the initial conditions ("other": `E4_DATA_SEED=100`; `experiments/results/replications/e4_indep`); five seeds each, held-out η, notation of Table 6.1.

Variant	ED / floor, main	ED / floor, other	magnetic balance, main	magnetic balance, other	null points, main	null points, other	restart drift E_B, main	restart drift E_B, other
W0 free components	236(6)	221(8)	0.159(15)	0.162(12)	1.11(5)	0.98(7)	0.513(9)	0.520(10)
W1 divergence penalty	464(12)	420(7)	0.410(5)	0.395(6)	0.698(12)	0.686(7)	0.5008(14)	0.4995(20)
W2 final projection	179(8)	169(9)	0.159(15)	0.162(12)	0.91(5)	0.79(8)	0.443(14)	0.453(16)
W3 projected flow	177(7)	164(8)	0.161(12)	0.165(13)	0.90(7)	0.82(8)	0.448(17)	0.452(19)
W4 vector potentials	256(7)	237(6)	0.149(14)	0.153(18)	1.14(7)	1.05(8)	0.517(10)	0.517(10)
W4s potentials, B → -B equivariant	209.8(19)	191(3)	0.264(14)	0.256(12)	1.83(7)	1.68(6)	0.382(12)	0.385(12)
W5 stationarity loss	215.0(19)	196.7(8)	0.516(13)	0.516(8)	2.36(2)	2.209(17)	0.334(3)	0.334(2)
W5 control (same fine-tuning, no identity)	210(2)	192(3)	0.251(15)	0.244(12)	1.79(7)	1.64(6)	0.390(12)	0.392(11)
W6 diffusion (VP, v-prediction)	519(17)	491(10)	0.213(11)	0.212(12)	0.43(2)	0.456(19)	0.637(2)	0.632(2)
W7 exponential tilting	195(11)	214(6)	0.43(4)	0.41(4)	2.24(19)	2.05(6)	–	–
W8 per-sample balance rescaling	117(2)	110(3)	0.195(19)	0.191(17)	0.71(8)	0.63(7)	–	–
W4c potentials, smooth condition	215.4(3)	205.8(7)	0.3843(16)	0.387(3)	2.142(12)	2.018(12)	0.3826(11)	0.3807(16)
G0 Gaussian per mode (no network)	3.52(7)	2.90(5)	0.9981(9)	0.9984(8)	2.235(7)	2.102(7)	0.212(3)	0.210(3)
W9 mode-wise affine term	9.3(3)	8.20(9)	0.0553(6)	0.0506(6)	1.665(14)	1.551(8)	0.124(3)	0.122(4)
W10 anchored on the Gaussian field	43.6(2)	43.3(2)	0.3092(17)	0.3122(19)	2.464(16)	2.323(15)	0.122(2)	0.125(4)

Variant	ED / floor, main	ED / floor, other	magnetic balance, main	magnetic balance, other	null points, main	null points, other	restart drift E_B, main	restart drift E_B, other
W11 anchor + affine term	8.6(2)	7.28(13)	0.0598(6)	0.0547(7)	1.696(12)	1.565(8)	0.120(2)	0.121(3)

5.9 Objectives of the race

The objectives of the team were quantified on the first day, before any model was trained (`race/teams/claude/objectives.toml`), and are measured by `experiments/race_metrics.py` on one generator, chosen on the validation values (Table 5.4).

Generator measured by C1–C5, T1 and T2: `W9_skip_sym`, the variant with the lowest energy-distance ratio on the validation η (file `experiments/results/e4_selection.json`).

Table 5.4: Objectives of the race against the measured values (generated by running the command of each objective).

Objective	Target	Measured	Seeds	Met
C1-3d-generator	≤ 10	9.35	5	yes
C2-divergence	$\leq 1.0 \times 10^{-6}$	1.6×10^{-8}	5	yes
C3-energy-balance	≤ 0.05	0.0567	5	no
C4-spectra-helicity	≤ 0.15	0.341	5	no
C5-topology	≤ 0.2	1.66	5	no
C6-comparative-study	≥ 6	12	5	yes
C7-toolbox	≥ 6	13	0	yes
T1-restart-drift	≤ 0.05	0.124	5	no
T2-polarity-symmetry	≤ 0.03	0.0186	5	yes

Five of the nine objectives are met: the distance to the reference law (C1, below ten floors), the divergence (C2), the number of decided comparisons (C6), the toolbox (C7) and the polarity symmetry (T2) reach their targets; the energy balance (C3) misses its target narrowly; the spectra and helicity (C4), the topology (C5) and the restart drift (T1) miss theirs by factors of two to eight. The targets were set before any model existed and were not changed. Three caveats on this table. The selection between W9 and W11 on the validation values is inside the seed noise (paired difference -0.29 ± 0.60): the measured generator could as well have been W11, with the same outcome for every objective. The objective C1 is met inside the uncertainty of the floor: the ratio is 9.35 with the floor of the protocol and 9.96 with the floor averaged over 20 splits, and one seed of the five is above 10 with the first floor, two with the second. And the selected generator is the best *learned* one: the Gaussian baseline G0, which has the lowest ratio on the validation values, is not a candidate, because the objective written on the first day names the model family (conditional flow matching on Fourier modes); the line of code that excludes it was written after its results were known. G0 would meet seven of the nine objectives, missing the topology and the restart drift.

6 Comparative study

Table 6.1 is the comparison the offer asks for: one line per mechanism or model family, one column per metric, on the three held-out values of η, seed mean and standard error. Table 6.2 gives, for each registered hypothesis, the status after review. Both are generated from the result files and the register; the sortable version of the first is in the explorer.

Table 6.1: Comparative table of E4 (measured; `experiments/results/e4_mhd3d_s<k>.json`). ED / floor: energy distance to the reference over the noise floor, at the training and at the held-out η; div B: largest relative divergence of B; energy residual: absolute residual of the mean energy balance over the mean dissipation; magnetic balance: absolute residual of the mean magnetic energy balance (0: production equals Ohmic dissipation; 1: no production); spectrum of B: relative L^2 error; magnetic helicity and null points: relative errors of the means; restart drift: relative drift of the magnetic and of the kinetic energy after $\tau = 4$; polarity: asymmetry of the sign of the cross helicity. Lower is better in every column. Notation: 217(14) is the seed mean 217 with a standard error of 14 on its last digits; values below 10^{-3} are given without error. The tilted ensemble W7 is a weighted sample whose effective size is about one sample: its line describes that sample. No restart test for W7 and for the rescaled ensemble W8.

Variant	Acts through	ED / floor, training η	ED / floor	div B	energy residual	magnetic balance	spectrum of B	magnetic helicity	null points	restart drift E_B	restart drift E_u	polarity	Seeds
W0 free components	none	165.4(5)	236(6)	0.454(2)	0.314(11)	0.159(15)	2.37(13)	1.99(19)	1.11(5)	0.513(9)	0.413(9)	0.015(3)	5
W1 divergence penalty	loss	275(4)	464(12)	0.261(6)	0.613(18)	0.410(5)	23.0(17)	38(3)	0.698(12)	0.5008(14)	0.709(4)	0.020(4)	5
W2 final projection	sampling	106.9(6)	179(8)	3.1×10^{-8}	0.314(11)	0.159(15)	2.04(13)	1.99(19)	0.91(5)	0.443(14)	0.354(11)	0.015(4)	5
W3 projected flow	sampling	105.8(6)	177(7)	2.6×10^{-8}	0.312(16)	0.161(12)	2.00(18)	1.9(2)	0.90(7)	0.448(17)	0.355(16)	0.015(2)	5
W4 vector potentials	design	175.4(5)	256(7)	1.4×10^{-8}	0.408(12)	0.149(14)	2.45(19)	2.1(3)	1.14(7)	0.517(10)	0.425(12)	0.013(3)	5
W4s potentials, B $\to$ -B equivariant	design	166.3(3)	209.8(19)	1.4×10^{-8}	0.284(10)	0.264(14)	0.97(8)	0.45(9)	1.83(7)	0.382(12)	0.281(16)	0.013(3)	5
W5 stationarity loss	loss	165.1(5)	215.0(19)	1.4×10^{-8}	0.191(5)	0.516(13)	0.473(15)	0.693(9)	2.36(2)	0.334(3)	0.243(3)	0.013(2)	5
W5 control (same fine-tuning, no identity)	loss	166.4(3)	210(2)	1.4×10^{-8}	0.290(10)	0.251(15)	1.03(9)	0.50(11)	1.79(7)	0.390(12)	0.289(16)	0.011(2)	5
W6 diffusion (VP, v-prediction)	family	160.4(6)	519(17)	1.3×10^{-8}	0.747(14)	0.213(11)	15.8(16)	22(2)	0.43(2)	0.637(2)	0.616(14)	0.022(6)	5
W7 exponential tilting	sampling	161(9)	195(11)	1.4×10^{-8}	0.204(8)	0.43(4)	0.51(6)	0.41(11)	2.24(19)	–	–	0.47(3)	5
W8 per-sample balance rescaling	sampling	92.9(3)	117(2)	1.4×10^{-8}	$< 10^{-12}$	0.195(19)	0.86(8)	0.52(11)	0.71(8)	–	–	0.0134(18)	5
W4c potentials, smooth condition	design	175.1(8)	215.4(3)	1.4×10^{-8}	0.2694(6)	0.3843(16)	0.637(6)	0.330(5)	2.142(12)	0.3826(11)	0.252(4)	0.0139(19)	5
G0 Gaussian per mode (no network)	family	2.67(8)	3.52(7)	1.5×10^{-8}	0.0015(3)	0.9981(9)	0.024(3)	0.035(5)	2.235(7)	0.212(3)	0.026(5)	0.018(3)	5
W9 mode-wise affine term	design	5.51(9)	9.3(3)	1.5×10^{-8}	0.0567(17)	0.0553(6)	0.145(3)	0.196(4)	1.665(14)	0.124(3)	0.053(6)	0.019(4)	5
W10 anchored on the Gaussian field	design	36.37(10)	43.6(2)	1.5×10^{-8}	0.1458(9)	0.3092(17)	0.2081(19)	0.096(7)	2.464(16)	0.122(2)	0.092(4)	0.0135(9)	5
W11 anchor + affine term	design	5.52(13)	8.6(2)	1.5×10^{-8}	0.0515(9)	0.0598(6)	0.143(4)	0.193(5)	1.696(12)	0.120(2)	0.053(3)	0.0169(10)	5

Table 6.2: Hypotheses of E4, registered with a refutation criterion before the computation (`research/hypotheses.json`), with their status after the review of the `encadrant` role.

Id	Hypothesis (shortened)	Status	Verdict of the `encadrant` role (beginning)
H4	In E4 (3D), generating vector potentials (W4_potentials, constraint by model design) gives a lower energy-distance ratio on the held-out η than generating free components (W0_free).	refuted	REFUTED (encadrant, 2026-11-12): W0_free - W4_potentials = -20.05 ± 3.21 (-6.2 SE), W4 below W0 in 0 of 5 seeds, negative at each held-out η; the two CPU seeds agree (-26.9).
H5	In E4, the divergence penalty on the end-of-path prediction (W1_penalty, weight 10; constraint by a training loss) does not enforce the constraint: […]	supported	CONFIRMED (encadrant, 2026-11-12): relative divergence of B 0.206 against 0.380 (ratio 0.544, not below one tenth); W0_free - W1_penalty = -228.2 ± 12.7, 0 of 5: the penalty does not enforce the constraint and degrades the law.
H6	In E4, the projected flow (W3_projected_flow: projected noise and projection of every integration stage) gives a lower energy-distance ratio on the held-out η than the Leray projection of the final samples of the free model (W2_final_projection).	refuted	REFUTED (encadrant, 2026-11-12): W2_final_projection - W3_projected_flow = 2.69 ± 3.72 (0.72 SE), 4 of 5 seeds: no difference at 2 SE; the CPU seeds show the same absence of effect.
H7	In E4, fine-tuning the equivariant potentials model with the law-level stationarity loss (W5_stationarity: energy identity and magnetic-helicity identity $E[\langle j \cdot B \rangle] = 0$, imposed at every η of the range) lowers the absolute residual of the mean energy balance on […]	supported	CONFIRMED (encadrant, 2026-11-12): |energy residual| W5_control - W5_stationarity = 0.0988 ± 0.0108 (9.1 SE), 5 of 5 seeds (0.290 → 0.191); ED ratio W5/control = 1.023.
H8	In E4, imposing the stationarity identities at training time (W5_stationarity) gives a lower energy-distance ratio on the held-out η than imposing the same identities at sampling time by exponential tilting of the samples of the control model (W7_tilting: […]	dropped	NOT DISCRIMINATING (encadrant, 2026-11-12): the registered criterion is formally met (W7_tilting - W5_stationarity_half = 30.0 ± 10.1, 5 of 5), but ESS/N is 1/256 in 14 of 15 cells: the tilted law is one sample and its energy distance is not a measurement of a law; with an estimate of the extra bias of a weighted sample the sign flips (-13.9 ± 9.5). […]
H9	In E4, the B → -B equivariant velocity field (W4s_potentials_sym, constraint by model design) lowers the polarity asymmetry $\lvert P(\langle u \cdot B \rangle > 0) - 1/2 \rvert$ of the generated law on the held-out η compared with the same model without symmetrization (W4_potentials), without […]	refuted	REFUTED (encadrant, 2026-11-12) on its first clause: asymmetry W4 - W4s = 0.0004 ± 0.0058, 3 of 5 seeds, both at or below the binomial floor 0.0176: the test could not discriminate at n = 512. The second clause holds: ED ratio W4s/W4 = 0.819 (-46.5 ± 4.8, 0 of 5).
H10	In E4, conditional flow matching (W4_potentials: linear path, Heun, 100 network evaluations) gives a lower energy-distance ratio on the held-out η than a variance-preserving diffusion with ancestral sampling (W6_diffusion: […]	supported	CONFIRMED as registered (encadrant, 2026-11-12): W6_diffusion - W4_potentials = 262.4 ± 12.1, 5 of 5, positive at each held-out η. Caveat of the encadrant: at the training η the diffusion model is better than flow matching (-15.06 ± 0.90, 0 of 5): the result is about interpolation in η with the aliased condition embedding, not about the model family at fixed η.

Id	Hypothesis (shortened)	Status	Verdict of the `encadrant` role (beginning)
H11	In the E4 reference data, the magnetic helicity spectrum is bipolar at every η: H(k = 1) < 0 and Σ_{k ≥ 2} H(k) > 0, with a large-scale energy fraction E_B(k = 1)/E_B between 0.2 and 0.8, so that the stationarity identity E[⟨j·B⟩] = Σ k² H(k) = 0 is a […]	refuted	REFUTED (encadrant, 2026-11-12): recomputed from the raw snapshots at the 11 η, independently of the experiment code, every value equal to e4_reference.json. H(1) < 0 at 11 of 11 η (z from -19 to -58) and E_B(1)/E_B in [0.370, 0.438] hold; Σ_{k ≥ 2} H(k) > 0 by 2 SE fails at η = 0.0365 and 0.03309 (z = -1.4 and -1.2, sign undetermined) and at η = 0.03 (z = -5.5, negative). […]
H12	In E4, the magnetic-helicity identity E[⟨j·B⟩] = 0 constrains the generated magnetic law more than the energy identity: on the samples of W4s_potentials_sym at the held-out η its normalized residual is at least twice that of the energy identity, and tilting […]	refuted	REFUTED (encadrant, 2026-11-12): seed mean of r_b/r_a = 0.504, r_b > r_a in 0 of 5 seeds. The tilting clause is met (|err H_M| tilt a - tilt b = 0.238 ± 0.095, 5 of 5) against a degenerate comparator (tilt on the energy identity: ESS/N 0.0096); tilt b against no tilt gains 0.110 ± 0.049; the two CPU seeds do not agree in sign.
H13	In E4, the error of a generator on the number of magnetic null points follows its error on the magnetic Taylor wavenumber k_λ = (⟨|j|²⟩/⟨|B|²⟩)^{}1/2 as N ∝ k_λ³: […]	refuted	REFUTED on the slope (encadrant, 2026-11-12): slope 4.875, R^2 0.902 over 135 points (seed-bootstrap interval [4.83, 4.92]); CPU 4.96. Protocol check passed (reference nulls change by 0.6 to 3.6 % between 48^3 and 64^3).
H14	In E4, the relative divergence of the free model W0_free is a representation error of the network and not an integration error (Proposition 2, consequence (c)): […]	supported	CONFIRMED (encadrant, 2026-11-12): relative divergence of B of W0_free 0.3792 at 50 Heun steps and 0.3788 at 200, ratio 0.998 to 1.000 per seed, above 0.1: integration error is excluded; 'representation error' is the remaining reading, not a separate measurement.
H15	In E4, the error of the generators at the held-out η comes from the embedding of the condition: the Fourier embedding of the pre-race code has frequencies up to 8 cycles over the normalized parameter range, above the Nyquist frequency of the 6 training values […]	refuted	REFUTED (encadrant, 2026-11-13): ratio of the seed-mean ED ratios W4c/W4 = 215.4/256.3 = 0.84 on the held-out η, above the 0.5 of the criterion; the paired difference ED(W4) - ED(W4c) = 40.9 ± 6.8 (6.0 SE, 5 of 5 seeds) is real, and absent at the training η (0.34 ± 0.32): the embedding of the condition explains part of the interpolation error, not the failure.
H16	In E4, a conditional Gaussian model with one mean and one covariance per Fourier mode (G0_gaussian_modes: no network; moments estimated at the training η and interpolated linearly in log η, symmetrized under B → -B) has a lower energy-distance ratio on the […]	supported	CONFIRMED (encadrant, 2026-11-13): G0 3.52 against 177 to 519 for W0–W6 on the held-out η (W3 the lowest, 176.6); best of them per seed 153–194 against G0 3.36–3.70, 5 of 5 seeds.
H17	In E4, the over-dispersion of the flow-matching generators comes from the bottleneck of the network (width 512 for 5320 latent dimensions), which cannot represent the linear part of the velocity field: […]	supported	CONFIRMED (encadrant, 2026-11-13): ratio W9/W4s = 9.35/209.8 = 0.045, below 0.2; ED(W4s) - ED(W9) = 200.4 ± 2.2 (91 SE), 5 of 5 seeds; flow-matching loss of W9 on a training sample 1.18, below π/2. The factor is 22.4 at the held-out η and 30.2 at the training η.
H18	In E4, flow matching adds to the second-order baseline what a Gaussian law cannot hold: with the mode-wise affine term and a longer training (W9L_skip_long, 8 times the iterations of W9), the drift of the magnetic energy when the solver is restarted from the […]	registered	not reviewed yet

Id	Hypothesis (shortened)	Status	Verdict of the `encadrant` role (beginning)
H19	In E4, a flow-matching field anchored on the exact field of the mode-wise Gaussian law, v = v_Gauss(x, t, η) + network (W10_anchored: […]	refuted	REFUTED (encadrant, 2026-11-13): ED(W10) - ED(G0) = +40.0 ± 0.26 on the held-out η (first clause not met); restart drift of E_B: G0 - W10 = 0.090 ± 0.003, 5 of 5 (met); null-count error: G0 - W10 = -0.229 ± 0.014, 0 of 5 (not met).
H20	In E4, the anchored model fails (W10_anchored on CPU seeds 0 and 1: 45 times the floor against 3.3 for G0) because the network's own linear component perturbs the second-order statistics that the fixed Gaussian field had right. […]	refuted	REFUTED (encadrant, 2026-11-13) on its first clause: ED(W11) - ED(G0) = +5.08 ± 0.22 (24 SE), 0 of 5 seeds; the two other clauses are met: restart drift of E_B G0 - W11 = 0.092 ± 0.003 (5 of 5), null-count error G0 - W11 = 0.539 ± 0.014 (5 of 5). Reported with it: […]
H21	In E4, the over-dispersion of the flow-matching generators without an explicit linear term comes from the width of the network and not from the training budget: […]	registered	not reviewed yet
H22	In E4, the failures of the mode-wise Gaussian law G0 on the magnetic energy balance, on the restart drift of the magnetic energy and on the number of null points come from the independence of its Fourier modes and not from its Gaussianity: […]	registered	not reviewed yet
H23	In E4, in the regime where the generator is close to the reference law (the model with the affine term, W9_skip_sym), fine-tuning with the law-level stationarity loss on the energy and magnetic-helicity identities (W12_skip_stationarity: […]	registered	not reviewed yet
H24	In E4, exponential tilting on the energy and magnetic-helicity identities does not collapse when the generated law is close to the reference: […]	registered	not reviewed yet

Reading the table by the place of the constraint:

- **By design.** Exact, and free at sampling time: the divergence is at round-off for every model on potentials. The design choice that changes the law is not the potential representation (W4 against W0) or the equivariance (W4s against W4, a gain of 46 floors out of 256) but the explicit linear part (W9, W11): a factor of twenty.
- **By the loss.** The divergence penalty does not enforce its constraint and costs accuracy. The law-level stationarity loss lowers the residual of the energy identity (from 0.29 to 0.19 against its control), at parameter values without data; it does not lower the residual of the magnetic-helicity identity imposed with it, and it changes other statistics in both directions (a better magnetic spectrum, a worse magnetic energy balance, more null points) without changing the distance to the reference law.
- **By the sampling procedure.** Projections give exact zero divergence and the best laws among the trained models of the first protocol; projecting every stage brings nothing over projecting the final sample. The tilting collapses on a few samples when the generated law is far from the identities. The per-sample rescaling satisfies the balance exactly, by removing the fluctuations of the energy budget that the true law has.
- **Model families.** At equal cost, flow matching interpolates in η better than the diffusion model; at the training values the two are equivalent. The Gaussian law is the baseline to beat on second-order statistics, and it is not beaten here; on everything that involves

third-order correlations or the dynamics (magnetic energy balance, null points, restart of the magnetic energy) the flow-matching models with an affine term are ahead of it.

7 Toolbox

The offer asks for "reproducible code integrated into the project toolbox". The toolbox of this repository is the Python package `src/milady/`; what this work added to it is drawn in the code-backed diagram code-architecture.html (Figure 7.1) and listed in Table 7.1.

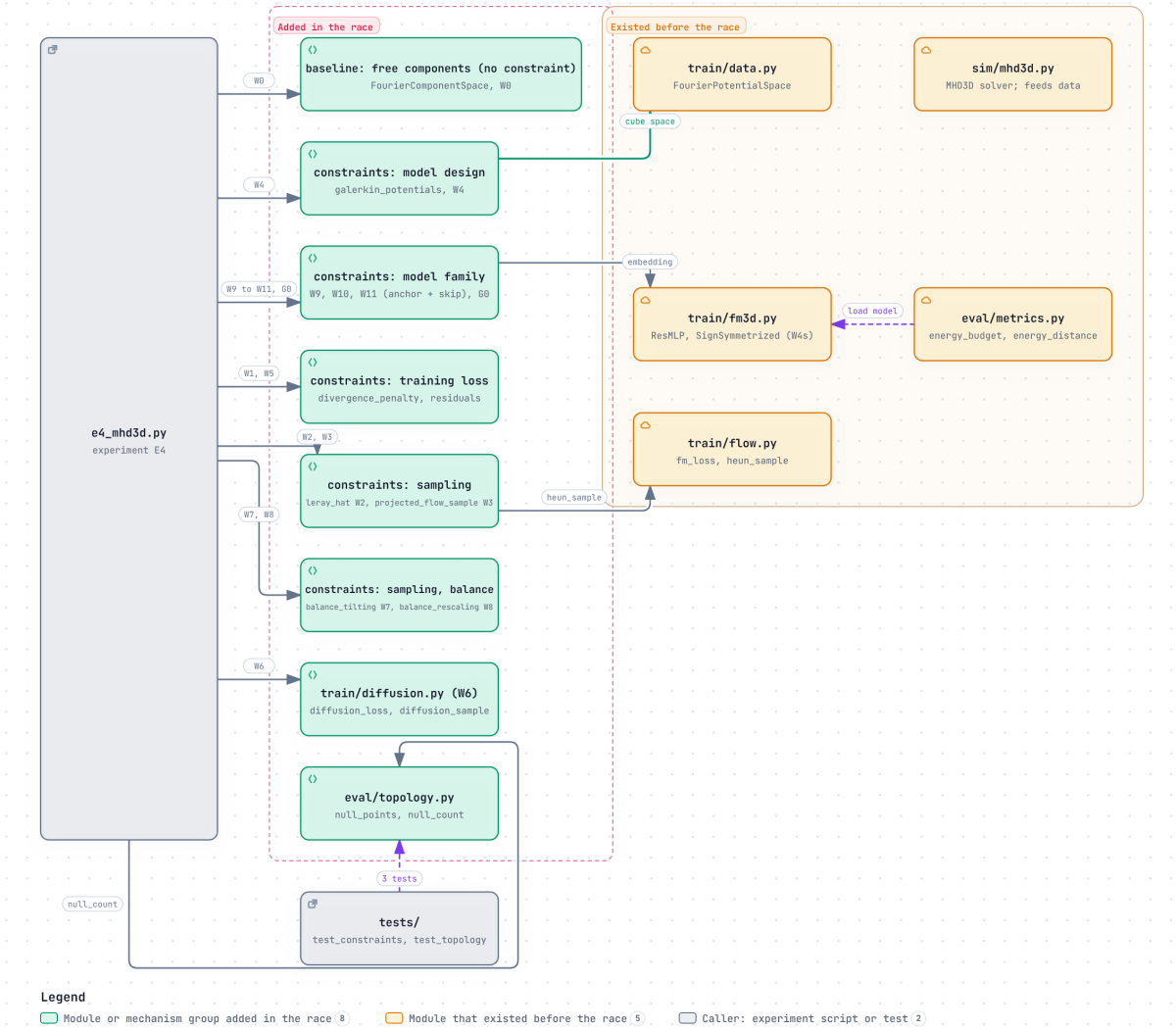


Figure 7.1: Code architecture of what E4 added to `src/milady/`: the constraint mechanisms by group, the diffusion module, the topology module, and the modules they build on. Static export of the Archify diagram `diagrams/code-architecture.html`.

Table 7.1: What entered the toolbox, with its tests.

Module	What it provides	Tests
`milady.train .constraints`	`REGISTRY` of 12 mechanisms: `FourierComponentSpace` (free components), `leray_hat`, `projected_flow_sample`, `divergence_penalty`, `budget_terms_hat` and `stationarity_residuals` (identities), `half_batch_product` (unbiased squared mean), `balance_tilting`, `balance_rescaling`, `ModewiseAffineSkip`, `ModewiseGaussian`, `GaussianAnchoredField`	`tests/ test_constraints.py`: one test per mechanism (13 tests)
`milady.train .data`	`FourierPotentialSpace` with `truncation="cube"`: potentials on exactly the modes of the solver	`test_galerkin_potentials` in the same file
`milady.train .diffusion`	`diffusion_loss` (v-prediction, variance-preserving path), `diffusion_sample` (ancestral or deterministic)	`test_diffusion` in the same file
`milady.train .fm3d`	`ResMLP` and `ConditionEmbedding` with a `cond_scale` argument (largest frequency of the embedding of the condition)	existing tests
`milady.eval .topology`	`null_points`, `null_count`: null points of a periodic 3D field by the trilinear method	`tests/ test_topology.py` (3 tests, with the Poincaré–Hopf balance of signs)
`tools/ runpod_job.py`	`run`, `collect`, `exec`, `cleanup`, `spend`: GPU jobs with a time limit, a ledger and per-team pod names	`tests/ test_pilot_tools.py`

How to call it, for a latent space `space` and a trained field `model`:

```
from milady.train import constraints as C
from milady.train.data import FourierPotentialSpace
from milady.eval.topology import null_count

# latent space: 665 modes, 8 channels (vector potentials of u and B)
space = FourierPotentialSpace(n=16, kmax=5, truncation="cube")
fh = space.field_hat(space.denormalize(x))      # (batch, 2 fields, modes, 3)
# stationarity identities: energy, magnetic helicity, cross helicity
terms = C.budget_terms_hat(space, fh, nu=0.05, eta=0.05, forcing_hat=f_hat)
res = C.stationarity_residuals(terms, nu=0.05, eta=0.05)
# unbiased estimate of (E<j.B>)^2 over 4 conditions
loss = C.half_batch_product(*res["magnetic_helicity"], n_groups=4)
# baseline G0 and the explicit affine term of W9
gauss = C.ModewiseGaussian(sign=space.sign).fit(x_train, c_train)
field = C.ModewiseAffineSkip(model, space.n_modes, space.n_channels, n_cond=1)
```

```
# null points of fields of shape (n, nx, ny, nz, 3)
counts = null_count(b_grid)
```

The tests check each mechanism against an independent computation: the identities against the time derivative of the solver, the Gaussian anchor against the closed form of Equation 3.4, the tilting against its moment conditions, the null finder against fields with a known number of nulls. `make test` runs them (the test suite, the linter and the check of the generated agent files pass on the branch).

The experiment itself is driven by scripts in `experiments/`: `e4_data.py` (data), `e4_mhd3d.py` (variants, by group), `e4_floor.py`, `e4_digest.py` and `digest.py` (digest), `race_metrics.py` (objectives), `e4_plots.py` (the data plots of this report), `e4_report_tables.py` (its tables) and `export_explorer.py` (the data of the explorer).

8 Discussion

8.1 What each place of constraint buys

The three places named by the offer do not play the same role, and the measurements suggest a division of labour.

Design is the place for what is exactly known. A linear constraint that holds for every sample (zero divergence), an exact symmetry of the law ($B \to -B$), and the second-order structure of a homogeneous field (the mode-wise affine part of the velocity field) can all be built into the model. They cost nothing at sampling time and they cannot be violated. The first two did not change the quality of the law in this experiment; the third decided it.

The loss is the place for what is known about the law and not about each sample. The stationarity identities are of this kind. Imposed at the level of the law, with an unbiased estimator, the energy identity moved the residual it targets, at parameter values without data; the magnetic-helicity identity did not, and the term changed statistics it was not aimed at. The identities are necessary conditions, not a substitute for a model that fits the data: a model a hundred floors from the reference stays there, and a loss on one moment of a wrong law can only trade one error for another.

The sampling procedure is the place for linear per-sample constraints, and a dangerous place for the others. Projection onto a linear constraint is harmless. Enforcing a law-level identity on each sample is a different operation: it produced the best-looking energy balance of the comparison and a law without the fluctuations of the real one.

Two results go beyond the mechanisms. The first is methodological: a baseline without a network, cheap to compute, was fifty times closer to the reference than the best trained model of the first protocol, and without it the comparison of constraints would have ranked models that were all wrong for the same reason. The second concerns evaluation: the law that is closest to the reference in a distance built on spectra, helicities and energies is a Gaussian law that is not a dynamo at all, since its mean magnetic production is zero. The magnetic energy balance, the null points and the restart test rank the models differently from the energy distance, and a comparative study that reported only one of the two would mislead. This is also a lesson for the choice of objectives: the objectives of this pilot, written on the first day, rewarded the energy distance and did not include the magnetic energy balance, which the review of the results showed to be the most discriminating physical metric.

8.2 Negative results and refuted claims

The register keeps what failed. Of the eleven hypotheses of the first protocol, written before any model was trained, six were refuted and one was dropped.

- *Potentials are not closer to the reference than free components* (H4 refuted). The constraint by design gives exact zero divergence at no cost, but the law is not better for it; the theory brief already says that with a perfect field the free model, the projections and the potentials give the same law, so any difference is an approximation effect, and here it has the opposite sign to the one expected.
- *The projected flow is not better than a final projection* (H6 refuted).
- *Equivariance does not lower the polarity asymmetry* (H9 refuted): the asymmetry was already at the sampling noise.
- *The helicity spectrum is not bipolar at* $k = 1$ *against* $k \geq 2$ (H11 refuted): the shell $k = 2$ is negative.
- *The magnetic-helicity identity is not the more violated one* (H12 refuted).
- *The null count does not scale as* k_λ^3 (H13 refuted on the slope).
- *Loss against tilting* (H8) was dropped: the criterion could be met by the sampling noise of one weighted ensemble.

Two claims made during the work were corrected by the reviewing role and are reported as corrected. The first version of the objective on the energy distance said that a perfect generator would give a ratio of one; with a floor computed from trajectory halves it would give about 0.6, and the note was corrected before any model result existed. The statement "the cause of the failure is the width of the network" was refused as a conclusion from the first runs and registered as a hypothesis with a width scan (H21, Section 5.6).

A run labelled as a replication was not one. The job `e4_main` reproduced the first GPU run bit for bit, because the data and model seeds were the same and the hardware identical; it is a reproducibility check. The replication with other data seeds is the one of Section 5.8.

8.3 Limits

- **Scale.** The system has 665 Fourier modes and $Re = 10$. The flow is smooth, the statistics of the Fourier modes are close to Gaussian at second order, and that is why a mode-wise Gaussian law is such a strong baseline here. Nothing guarantees this at the Reynolds numbers of the MilaDy simulations, where the non-Gaussian part of the law carries the physics (intermittency, coherent structures). The ranking of the mechanisms may change with scale; the method of comparison does not.
- **Architecture.** The network acts on the flattened vector of modes. This does not scale: at 64^3 the latent has more than 10^5 dimensions. The explicit affine term is mode-wise and scales linearly with the number of modes, but the nonlinear part needs an architecture with locality in Fourier or physical space (a convolutional or attention-based field, as in the references of the section State of the art).
- **Homogeneity.** The mode-wise form of the affine term and of the Gaussian baseline G0 treats different Fourier modes as uncorrelated, which is true of a statistically homogeneous field. It is not true here: the forcing has a fixed phase, the mean flow is strong, and the covariance between modes carries the mean magnetic production (Section 3.2). In a bounded domain (the von Kármán geometry of the project) the departure from homogeneity is larger still, and the natural basis is different.
- **Metrics.** The energy distance is computed on 21 physical features, all second-order quantities of one snapshot. A law can be close to the reference in this distance and wrong in its higher-order structure; the null points and the restart test are there to see that, and they do.

- **One conditioning parameter.** Only η varies, over a range in which the dynamo is always saturated and the law changes smoothly. The dynamo threshold, where the law changes fastest (Proposition 9 of the theory brief), is outside the range.
- **Statistics.** Five seeds per variant; one realisation of the reference data for the main run and one independent realisation for the replication. The objective on the distance to the reference law is met inside the uncertainty of the noise floor.
- **Restart test.** The drift is a transient that is not monotonic in time; the metric of the objectives measures it at one time, $\tau = 4$. The last runs record it at four times.

8.4 What the incidents of the pilot teach

Four incidents cost time and are part of the result, because the race is also a test of the method. Rented GPUs were refused for lack of capacity for about an hour. The first pod ran for 24 minutes without its status being read, because a proxy refused the default user agent of the HTTP client. Another session sharing the cloud account deleted two running pods with its clean-up command. The container of the session restarted twice and orphaned running jobs. Each was repaired in the shared tool with a test (user agent, per-team pod names, the `collect` command), and announced in the journal. One bug was in the experiment script itself: a loop variable overwrote the suffix of the result files after a certain variant, and one run brought back incomplete files; it was found by reading what a job returned instead of trusting its exit code, fixed, and covered by a static test.

9 Conclusion

This report compared, on a small parametric MHD dynamo, the three places where a physical constraint can enter a generative model. All statements below are **measured** on this system, with five seeds, on parameter values that were not used for training.

1. Zero divergence is obtained exactly by design (potentials) or by projection at sampling time; a penalty in the loss does not enforce it and degrades the law.
2. None of these divergence mechanisms decides the quality of the generated law. What decides it is whether the velocity field of the flow-matching model can carry the linear, second-order part of the transport: an explicit mode-wise affine term brings the same network from about 210 noise floors to about 9.
3. A Gaussian law per Fourier mode, without any network, is closer still in energy distance (3.5 floors), but it is not a dynamo: its mean magnetic production is zero, so its magnetic energy balance is violated completely, it has more than three times the reference number of null points, and the magnetic energy drifts when the solver restarts from its samples. The flow-matching models with the affine term bring the residual of the magnetic balance from 1 to 0.06 and halve that drift.
4. A law-level stationarity loss can be imposed at parameter values without data. On a model far from the reference it lowers the residual of the energy identity by a third, leaves the residual of the parameter-free identity $\mathbb{E}[\langle j \cdot B \rangle] = 0$ unchanged, and has side effects on other metrics. Imposing a balance on each sample satisfies it exactly and removes real fluctuations.
5. No generator reproduces the magnetic topology: the models closest to the reference law have about three times the reference number of null points.

What remains **conjectured**: that the failure of the plain networks is a matter of width (the scan of the section Results tests it), and that the ranking of the mechanisms carries over to turbulent regimes, where the non-Gaussian part of the law dominates.

The next steps for the internship follow from the limits: an architecture with locality for the nonlinear part of the field, keeping the explicit affine term; the same protocol on a 64^3 box at higher Reynolds numbers, where the solver of the toolbox runs at a few milliseconds per step on a GPU; a range of η that crosses the dynamo threshold; and topology-aware training signals, since topology is where every model fails.

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11 Reproducibility

11.1 Commands

Everything is in the branch `claude/modest-newton-zx8fqs` of the repository. From the root:

```
make test                                    # tests of src/milady and of the tools
cd experiments
python3 e4_data.py --quick                   # reference data (--jobs N: full data)
python3 e4_mhd3d.py --reference --quick      # reference statistics and floors
python3 e4_mhd3d.py --quick --seed 0 --group all   # groups A-E (L: long; S: scan)
python3 e4_floor.py                          # noise floors
python3 digest.py > results/digest.md        # digest of E1-E4
python3 race_metrics.py --all                # objectives of the race
python3 e4_plots.py                          # data plots of this report
python3 e4_report_tables.py                  # tables of this report
python3 export_explorer.py --team claude     # data of the explorer
# on a GPU pod (5 seeds), from the root:
python3 tools/runpod_job.py run --hours 2 \
    --outputs experiments/results/replications/e4_final \
    -- bash experiments/e4_gpu_job.sh full "0 1 2 3 4" e4_final all
python3 tools/render.py teams/claude/report teams/claude/explorer
python3 tools/check_deliverables.py --team claude --built
```

Without `--quick`, the data take about 25 minutes on 11 CPU cores and one seed of the groups A to E about 35 minutes on a GPU. Quick outputs are not versioned and are never quoted.

11.2 Result files and commits

Result set	Location	Code commit of the run	Commit of the results
Objectives of the team (day 1)	`race/teams/claude/objectives.toml`		`90e64af`
Register: H4–H14, H15, H16–H18, H19–H20, H21	`research/hypotheses.json`	–	`68e2fd4`, `9fd7272`, `fc04247`, `f3868f5`, `5027f7c`
Main run: groups A–E, 5 seeds (GPU job `e4_final`)	`experiments/results/e4_*.json`	`973fe3a`	`991d682`

Result set	Location	Code commit of the run	Commit of the results
First GPU run of groups A and B (reviewed for H4–H14)	`experiments/results/replications/e4_gpu`	`d0e018f`	`fb0786b`
Bitwise re-run of groups A and B	`experiments/results/replications/e4_main`	`62e81ca`	`8184bc9`
Convergence of the data	`experiments/results/replications/e4_convergence`	`d0e018f`	`56dbb91`
CPU seeds 0 and 1 (container)	`experiments/results/replications/e4_cpu`	`5e4ca1d`	`991d682`

: Result sets, with the commit of the code that produced them and the commit that added them. The registrations precede the results of the hypotheses they concern. {#tbl-commits}

The main run was launched with a file-naming bug (the section Discussion): its result files were written under another suffix, copied back by a loop started on the pod, and renamed; the `encadrant` checked the renamed files against the per-seed logs of the job (`research/gpu-logs/e4-final/`).

11.3 GPU jobs and costs

Every GPU job of the pilot is in `research/compute-ledger.jsonl` (Table 11.2, Figure 11.1); the pods were rented on demand and deleted at the end of each job. Jobs with a non-zero exit code are the capacity failures, the interrupted jobs and the incidents described in the section Discussion; the two negative entries are corrections made after another session deleted running pods, so that the ledger matches what was billed.

Total: 22 ledger entries, 10.61 GPU hours, 11.04 USD.

Table 11.2: GPU jobs of the pilot (generated from `research/compute-ledger.jsonl`).

End (UTC)	GPU	USD/h	Hours	USD	Exit	Commit	Command (shortened)
11-12 17:48	RTX PRO 4500 Blackwell	0.72	0.404	0.29	–	`36908cc`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 17:58	RTX PRO 4500 Blackwell	0.72	0.026	0.02	–	`40f6188`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 17:59	RTX PRO 4500 Blackwell	0.72	0.026	0.02	–	`40f6188`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 18:18	RTX PRO 4500 Blackwell	0.72	0.104	0.07	–	`d4f2135`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 18:25	RTX PRO 4500 Blackwell	0.72	0.026	0.02	0	`f099fa8`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 18:27	RTX PRO 4500 Blackwell	0.72	0.026	0.02	0	`f099fa8`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 18:40	RTX A6000	0.69	0.026	0.02	0	`47d057d`	python -m milady.sim.mhd3d bench –n 64 128 –device cuda – …
11-12 18:44	RTX PRO 4500 Blackwell	0.72	0.146	0.11	0	`8fd79a0`	e4_gpu_job.sh quick 0
11-12 19:03	RTX PRO 4500 Blackwell	0.72	0.172	0.12	0	`d0e018f`	e4_convergence_job.sh

End (UTC)	GPU	USD/h	Hours	USD	Exit	Commit	Command (shortened)
11-12 20:16	RTX PRO 4500 Blackwell	0.72	1.415	1.02	0	`d0e018f`	e4_gpu_job.sh full “0 1 2 3 4”
11-12 22:21	RTX PRO 4500 Blackwell	0.72	0.590	0.42	–	`1c21afc`	e4_gpu_job.sh full “0 1 2 3 4” e4_gpu_de D,E
11-12 22:23	RTX PRO 4500 Blackwell	0.72	1.820	1.31	–	`a20bcb1`	e4_gpu_job.sh full “0 1 2 3 4” e4_gpu_all
11-12 22:24	RTX PRO 4500 Blackwell	0.72	0.031	0.02	–	`837c23a`	e4_gpu_job.sh full “0 1 2 3 4” e4_main all
11-12 22:24	RTX PRO 4500 Blackwell	0.72	-0.590	-0.42	–	`1c21afc`	CORRECTION of the entry of pod hd6qx71mm6jekm: the pod was d …
11-12 22:24	RTX PRO 4500 Blackwell	0.72	-0.628	-0.45	–	`a20bcb1`	CORRECTION of the entry of pod 00t6s3l481flvm: the pod was d …
11-12 23:24	A100-SXM4-80GB	2.19	0.126	0.28	–	`5027f7c`	e4_gpu_job.sh full “0 1 2 3 4” e4_scan S
11-12 23:24	RTX PRO 4500 Blackwell	0.72	0.126	0.09	–	`5027f7c`	e4_gpu_job.sh full “0 1 2 3 4” e4_long L
11-12 23:27	RTX PRO 4500 Blackwell	0.72	0.983	0.71	0	`62e31c9`	collected after a restart of the session: milady-claude-job- …
11-13 00:17	RTX PRO 4500 Blackwell	0.72	0.879	0.63	0	`973fe3a`	collected after a restart of the session: milady-claude-job- …
11-13 00:55	RTX PRO 4500 Blackwell	0.72	1.508	1.09	0	`973fe3a`	collected after a restart of the session: milady-claude-job- …
11-13 01:25	GeForce RTX 4090	0.89	1.356	1.21	0	`dfc02a2`	env E4_DATA_SEED=100 e4_gpu_job.sh full “0 1 2 3 4” e4_indep …
11-13 01:27	A100-SXM4-80GB	2.19	2.034	4.46	124	`973fe3a`	collected after a restart of the session: milady-claude-job- …

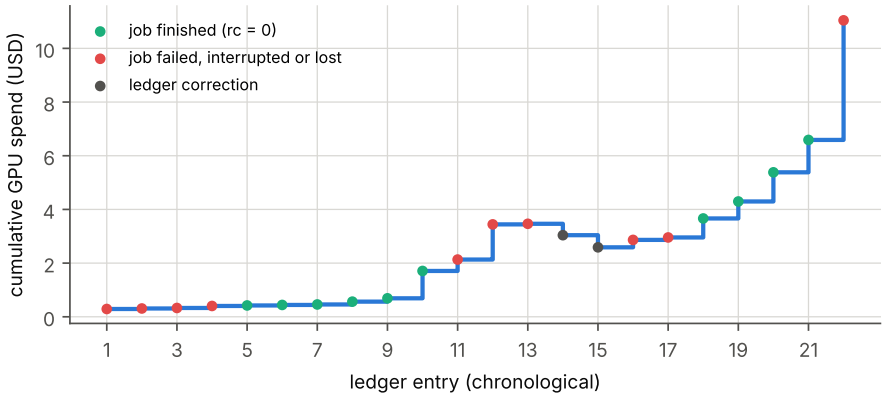


Figure 11.1: Cumulative GPU spend of the pilot, one point per entry of the compute ledger. Data plot from `research/compute-ledger.jsonl`.

The cost of the session itself (tokens, sub-agent calls, CPU time) is reported by `tools/usage_report.py` in `research/usage/latest.md`.

11.4 Figures of this report

The eight schematics were drawn with the `schematic-figure` skill (sources in `site/teams/claude/figures/<n` rendered to SVG and PDF), the three diagrams with Archify (`site/teams/claude/diagrams/`),

and the data plots by `experiments/e4_plots.py` from the result files named in each caption (`site/teams/claude/plots/plots.json` lists them).